PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

Smart Tourist Safety Monitoring & Incident Response System using Al ,Geo-Fencing, and Blockchain-based Digital ID

A PROJECT REPORT

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IN

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report "Smart Tourist Safety Monitoring & Incident Response System using Al ,Geo-Fencing, and Blockchain-based Digital ID" is a bonafide work of "N SAI SANDEEP REDDY (20221CSE0277), HARSHITH K R(20221CSE0203) & SOHAN K (20221CSE0418)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING during 2025-26.

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DECLARATION

We, the students of final year B.Tech in COMPUTER SCIENCE ENGINEERING at Presidency University, Bengaluru, named N SAI SANDEEP REDDY, HARSHITH K R & SOHAN K, hereby declare that the project work titled **"Smart Tourist Safety Monitoring & Incident Response System using Al ,Geo-Fencing, and Blockchain-based Digital ID"** has been independently carried out by us and submitted in partial fulfilment for the award of the degree of B.Tech in COMPUTER SCIENCE & ENGINEERING during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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Abstract

The high rate of development in global tourism has led to the increasing demand of effective systems that allocate safety and security of the travelers of the global world especially in environments that are unfamiliar to them. The complications commonly experienced by the tourists include entering places of danger, emergency response, identity tests, and real-time tracking. Traditional tourist safety systems are largely manual and reactive and therefore do not correspond to the needs of smart city. Thus, it is highly demanded to have a smart, automated, and secure monitoring tool, which is capable of identifying the risk in time and supporting tourists in real-time. To overcome these difficulties, this project offers a Smart Tourist Safety Monitoring System which combines the application of Artificial Intelligence (AI), Geo-Fencing, and Digital ID based on Blockchain and is implemented on the web-based platform. The system keeps track of the location of the tourist through geo-fencing to make sure the user does not go anywhere that is not in a predetermined safe area. In case a tourist leaves a protective zone or goes into a dangerous zone, alerts are automatically issued by the system. A user behavior-based artificial intelligence module processes patterns of behaviour including abrupt changes of location and excessive idleness to calculate a dynamic risk indicator and detect potentially unsafe conditions. To secure the identity, the system computes a unique hash digital identity of the users, through a blockchain model, and supports the integrity and lack of tampering with the data, without revealing sensitive personal information. It also features secure Register and Login authentication, a Tourist Dashboard where the safety indicators display live and SOS and Admin Dashboard to track the alerts and respond to emergencies. Overall, the project demonstrates that intelligent monitoring system network can enhance tourist safety, allow responding to the emergency situation faster, and offer effective digital safety support network in the context of modern traveling.

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Abbreviations

Abbreviation	Full Form
API	Application Programming Interface
CSS	Cascading Style Sheets
Firebase	Firebase Real-time Database
GPS	Global Positioning System
GUI	Graphical User Interface
HTML	HyperText Markup Language
HTTPS	HyperText Transfer Protocol Secure
IST	Indian Standard Time
JSON	JavaScript Object Notation
JWT	JSON Web Token
QR	Quick Response
RBAC	Role Based Access Control
SOS	Save Our Souls
UI	User Interface
UX	User Experience

Chapter 1
INTRODUCTION

Tourism has become one of the fastest-growing sectors in the world today, playing an important role in connecting people, cultures, and economies. It not only contributes to a country's income but also creates job opportunities and supports local businesses, especially in countries like India. With the availability of affordable travel options and easy access to information through the internet, more people are now exploring new destinations than ever before. Traveling has become more than just a luxury; it is now a common part of modern life. However, along with this rapid growth, new challenges have also emerged, particularly in terms of safety and security. Tourists often visit unfamiliar places where they may not know the local language, culture, or environment, which can make them more vulnerable. Situations like theft, getting lost, accidents, or delays in receiving help can affect their overall travel experience. In many cases, the lack of immediate support and real-time monitoring systems makes these situations more difficult to manage. Because of this, ensuring the safety of tourists has become a major concern in today's tourism industry.

The integration of advanced technologies like Artificial Intelligence (AI), Geofencing, and Blockchain offers a promising solution to these challenges. This project, *"Smart Tourist Safety Monitoring & Incident Response System Using AI, Geo-Fencing and Blockchain-based Digital ID,"* is designed to develop a system that not only keeps track of tourist safety but also provides quick and effective responses during emergencies.

By combining intelligent monitoring with secure identity management and location-based alerts, the system focuses on both preventing incidents and responding to them efficiently. The goal is to make travel safer and give tourists more confidence while exploring new places.

In addition to improving safety, the system also aims to strengthen coordination between different groups involved in tourism, such as local authorities, emergency services, and support teams. By bringing everything onto a single platform where information can be shared quickly and securely, the system helps reduce delays during critical situations. This improves response time and ensures that help reaches tourists when they need it the most. Overall, the project envisions a smarter and

safer tourism environment where technology works quietly in the background to protect travelers without affecting their experience.

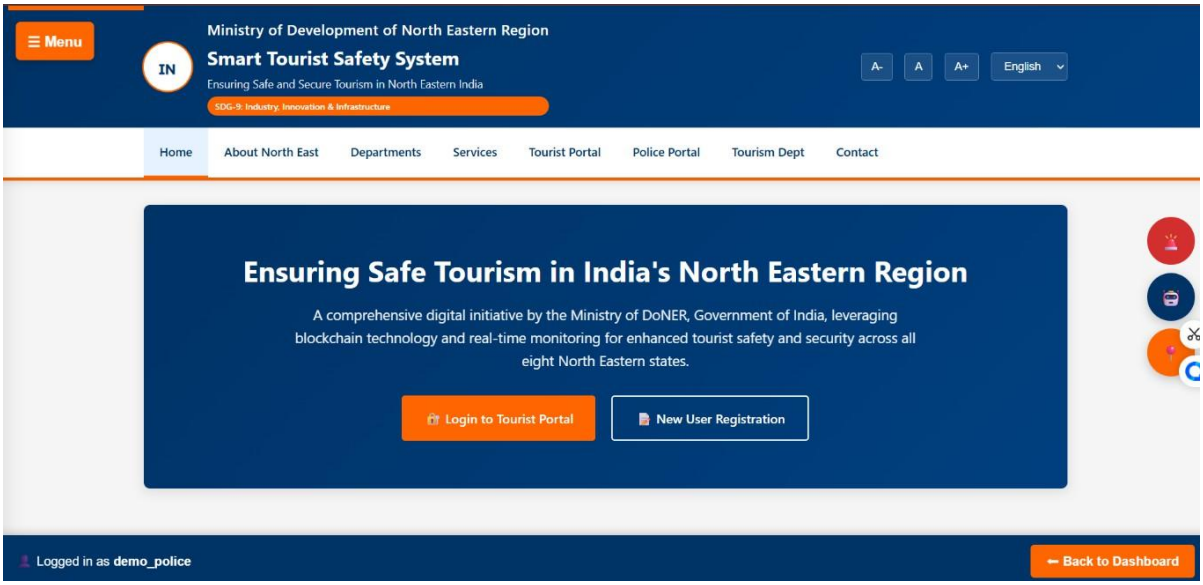

Fig 1.1 Architecture and Functional Overview of Smart Tourist Safety Monitoring & Incident Response System

1.1 Background

The rapid growth of tourism, especially in developing countries like India, has brought both exciting opportunities and noticeable challenges. On one side, tourism helps the economy grow, supports small businesses, and creates jobs for many people. On the other side, the increasing number of tourists has raised concerns about safety, communication issues, and how emergencies are handled. When people travel to new places, they are often not familiar with the local language, culture, or environment, which can make them more vulnerable to different kinds of risks.

In many situations, the safety systems that are currently in place are not fully effective in handling these challenges. Most traditional systems depend on manual work, slow communication, or separate technologies that do not work together properly. For example, a tourist might unknowingly enter an unsafe area because there is no proper alert system, or authorities may not have quick access to the tourist's identity details during an emergency. These gaps can delay the response and make the situation more difficult to manage.

With the growth of technologies like Artificial Intelligence, Geofencing, and Blockchain, better solutions are now possible. These technologies can help in analyzing data, tracking locations in real time, and storing identity information securely. When combined, they provide a strong base for building a smarter and more reliable tourist safety system, which is the main idea behind this project.

As tourism continues to grow, the need for a smarter and more connected safety system becomes even more important. A system that can bring together location tracking, quick decision-making, and secure identity verification in one place can make a big difference. It can help reduce response time, improve communication between authorities, and provide accurate information during emergencies. In the end, such a system not only makes travel safer for tourists but also helps create a sense of trust, encouraging more people to explore new places with confidence.

At the same time, such systems can also help government authorities and tourism departments manage things more effectively. By studying the collected data, they can understand which areas are more risky, how tourists move around, and where improvements are needed. This makes it easier to take preventive steps instead of reacting only after something goes wrong. Over time, this approach can lead to safer tourist destinations and better travel experiences for everyone, while also supporting the development of smarter and more organized cities.

1.2 Statistics of the Project

1.2.1 Growth of Global Tourism

Tourism is growing steadily around the world, with more people traveling every year, which increases the importance of safety systems.

- International tourists (2023) – 1.3 billion
- Annual tourism growth rate – 3–5%
- Domestic tourism share – 70–75%

1.2.2 Contribution of Tourism to Economy

Tourism continues to be an important part of the global economy, supporting jobs and overall development.

- Global GDP contribution – 9–10%
- Employment generated – 300+ million jobs
- India tourism GDP contribution – 5–6%

1.2.3 Tourist Safety Concerns

For most travelers, feeling safe has become one of the main factors when planning a trip.

- Tourists prioritizing safety – 60–70%
- Travelers avoiding unsafe destinations – 30%
- Solo travelers with safety concerns – 50%+

1.2.4 Crime and Incident Reports in Tourism

Incidents such as theft, scams, and getting lost are still common in many tourist locations.

- Petty crime cases (theft, scams) – 30–40%
- Tourist-related incidents annually – Thousands of cases
- Missing/lost tourist cases – Significant proportion

1.2.5 Emergency Response Delay

Quick response is very important during emergencies, but delays are still common in many situations.

- Average response time – 10–15 minutes
- Risk increase due to delay – up to 40%
- Response time reduction (smart systems) – 40–60%

1.2.6 Smartphone and Internet Usage

Today, most travelers depend on smartphones, which makes it easier to provide digital safety solutions.

- Global smartphone users – 5+ billion
- Tourists using mobile apps – 65–70%
- GPS/navigation usage – 80%+ travelers

1.2.7 Growth of Smart Technologies

Advanced technologies are developing quickly and are being used in many real-world applications.

- Artificial Intelligence market size (2030) – $1+ trillion
- Internet of Things devices (projected) – 29+ billion
- Blockchain adoption growth – 60%+ annually

1.2.8 Effectiveness of Geo-Fencing Systems

Geo-fencing helps in monitoring locations and sending alerts instantly when needed.

- Tracking efficiency improvement – ~40%
- Alert response time – Real-time (seconds)
- Usage in location-based apps – 50%+

1.3 Prior Existing Technologies

Before the development of integrated smart systems, different technologies were used individually to improve tourist safety. Each of these solutions was useful in its own way, but most of them worked separately and were not connected to each other. Because of this lack of coordination, they were not always effective in handling real-time emergencies or providing complete safety support to tourists.

1.3.1 GPS Tracking Systems

GPS-based systems are commonly used to track the real-time location of individuals. They help in identifying where a person is at any given moment, which can be very helpful during emergencies or when someone gets lost. However, these systems mainly focus only on location tracking and do not provide smart features like automatic alerts or risk detection.

- Real-time location tracking – Provides current position of users
- Navigation support – Helps in route guidance
- Lack of smart alerts – No automatic risk detection

1.3.2 Mobile Safety Applications

Mobile safety applications are designed to help users by offering features such as emergency contacts, SOS buttons, and location sharing. These apps are easy to use and widely available. However, in most cases, they depend on the user to manually trigger alerts, which may not always be possible during stressful or dangerous situations.

- SOS alert feature – Sends emergency signals to contacts
- Location sharing – Shares live location with others
- Manual activation required – Needs user action during emergencies

1.3.3 Emergency Helplines

Emergency helplines provide a direct way for tourists to contact support services when they face problems. They play an important role in offering assistance, especially during serious situations. However, response times can sometimes be slow, and these systems often do not have enough real-time information about the tourist's exact location or condition.

1.3.4 QR Code-Based Identification Systems

QR code-based systems are used to quickly access user information by scanning a code through a mobile device. They are simple, cost-effective, and widely used in many applications such as ticketing and identification. However, traditional QR codes are not highly secure and can be copied or misused if not properly protected.

- Quick data access – Information retrieved through scanning
- Easy to use – Works with smartphones
- Low security in basic systems – Can be copied or misused

1.3.5 IoT-Based Monitoring Systems

Internet of Things (IoT)-based systems use connected devices and sensors to collect and monitor data in real time. These systems can track movements and detect certain conditions, which can be helpful for safety purposes. However, they often lack advanced decision-making capabilities and may require complex setup and maintenance.

1.3.6 Location-Based Services (LBS)

Location-Based Services use mobile networks and GPS to provide services based on a user's current location, such as navigation, maps, and nearby recommendations. While these services are very helpful for general travel guidance, they are not specifically designed for safety and do not provide real-time emergency response features.

1.4 Proposed Approach

The proposed system is designed to improve tourist safety by bringing together different technologies into one connected platform. It focuses on providing continuous monitoring, quick alerts, and fast emergency response. Instead of depending on separate systems, everything works together smoothly to handle situations more effectively. This approach helps reduce risks and ensures that tourists feel safer while traveling.

1.4.1 Monitoring System

The system continuously observes user activity and identifies unusual situations to improve safety.

- Core Functions
    - Detects unusual behavior
    - Identifies possible risks

    - Sends alerts automatically
- Benefits
    - Reduces manual effort
    - Supports early warning

1.4.2 Geo-Fencing Technology

Geofencing is used to create virtual boundaries to monitor safe and restricted areas.

- Zone Management
    - Defines safe and restricted areas
- Alert System
    - Detects entry/exit
    - Sends instant alerts

1.4.3 Blockchain-Based Digital Identity

The system uses Blockchain to store identity data in a secure and reliable way.

- Security
    - Tamper-proof data
- Usage
    - Quick verification
    - Maintains privacy

1.4.4 QR Code Integration for Identity Access

Each tourist is provided with a QR code that is linked to their identity for easy access.

- QR Code
    - Generated during registration
- Usage
    - Quick scanning

- Easy access in emergencies

1.4.5 Real-Time Incident Response System

This system helps tourists quickly connect with authorities during emergencies.

- Communication
  - Instant alerts
  - Direct contact with police
- Efficiency
  - Faster response

1.4.6 Incident Image Capture and Reporting

The system allows tourists to capture and send images of incidents to authorities.

- Capture
  - Take photos in real time
- Reporting
  - Send to police instantly
  - Helps quick action

1.5 Objectives

The main objective of this project is to develop a smart system that improves tourist safety and ensures quick support during emergencies.

- To design a system that improves tourist safety by connecting different technologies in one platform.
- To provide real-time location tracking so that tourists can be monitored continuously.
- To implement Geofencing to identify unsafe movements and restricted areas.
- To develop a secure Blockchain-based identity system for safe data storage.
- To provide quick and efficient response during emergency situations.

- To reduce dependency on manual monitoring by using automated processes.
- To enable QR code-based identity access for faster verification.
- To generate real-time alerts to keep tourists informed about risks.
- To improve communication between tourists and authorities for faster help.
- To allow tourists to report incidents by sending images.
- To improve overall safety by identifying and reducing possible risks.
- To build trust and confidence among tourists while traveling.

1.6 Sustainable Development Goals (SDGs)

This project supports several Sustainable Development Goals by improving safety, strengthening public systems, and supporting tourism development. It helps in creating a safer and more reliable environment for travelers while also contributing to overall social and economic growth.

1.6.1 Industry, Innovation and Infrastructure (SDG 9)

- Promotes use of modern technologies in safety systems
- Supports development of smart infrastructure

1.6.2 Sustainable Cities and Communities (SDG 11)

Safer Public Spaces The system helps improve safety in tourist areas by monitoring activities and reducing risks in crowded or unfamiliar locations. This creates a more secure environment for both tourists and local residents. Real-time GPS tracking, geo-fencing alerts, and emergency SOS features ensure immediate response to any safety threat, making public spaces safer and more accessible for everyone. Better City Management It supports authorities in managing public spaces more effectively by providing real-time information, which helps in taking quick and informed decisions during critical situations. The admin and police dashboards provide live monitoring, incident tracking, and geo-fence zone management, enabling authorities to allocate resources efficiently, respond to emergencies promptly, and maintain overall public order in tourist-heavy urban areas.1.6.3 Peace, Justice and Strong Institutions (SDG 16)

- Strengthens coordination between tourists and authorities
- Supports faster and more organized emergency response

1.6.4 Decent Work and Economic Growth (SDG 8)

Tourism Growth Improved safety encourages more people to travel without fear, which increases tourist flow and promotes tourism development. The system's real-time monitoring, emergency SOS alerts, and blockchain-secured Digital ID verification build trust among tourists, motivating more domestic and international visitors to explore destinations confidently, ultimately boosting overall tourism growth across the region. Economic Benefits As tourism grows, it creates more job opportunities and supports local businesses, contributing to overall economic growth. By ensuring tourist safety through advanced technology, the system attracts higher tourist footfall, directly benefiting local hotels, restaurants, transport services, and small businesses. Increased tourism revenue strengthens the local economy and generates sustainable employment opportunities for the community.

1.6.5 Good Health and Well-Being (SDG 3)

- Helps reduce risks and unsafe situations for tourists
- Ensures quick support and assistance during emergencies

Fig 1.2 Mapping of Smart Tourist Safety Monitoring System to Sustainable Development (SDGs)

1.7 Overview of the Project

The Smart Tourist Safety Monitoring and Incident Response System is a web-based application developed to enhance the safety and security of tourists visiting destinations across India. The system provides a unified digital platform connecting tourists, police officers, and administrators through role-based dashboards, enabling real-time monitoring, emergency response, and incident management. The primary purpose of this system is to bridge the gap between tourists in distress and emergency responders. Traditional tourist safety mechanisms rely on manual reporting and delayed communication, resulting in slow response times during critical situations. This system addresses these limitations by providing instant digital communication, real-time location tracking, and automated alert systems. The system covers three major functional areas. The tourist module provides real-time GPS tracking, emergency SOS alerts, incident reporting with image evidence, and blockchain-secured Digital ID generation with QR code verification. The police module offers live tourist tracking on Google Maps, real-time SOS alert management, and comprehensive incident history tracking. The administrator module provides system-wide statistics, geo-fence zone management, and advanced analytics. A key feature of the system is the Geo-Fencing and Alert System, which visually represents Safe Zones in green, Restricted Areas in red, and Tourist Hubs in blue on an interactive Google Maps interface. The system uses Firebase Firestore as its real-time cloud database, ensuring instant data synchronization across all dashboards. The system is built using React.js, Firebase Authentication, Firebase Firestore, Google Maps API, and Tailwind CSS. It supports direct access to government tourism portals including Karnataka Tourism and the Government of India Tourism Portal. Overall, this system improves tourist safety, encourages confident travel, and supports better coordination between tourists and authorities, contributing to India's vision of safe and smart tourism development. The application is fully responsive, ensuring seamless usability across mobile devices, tablets, and desktop computers. It also displays live Indian Standard Time (IST), essential India emergency contact numbers, and real-time geo-fence alerts, making it a complete and reliable safety solution for tourists traveling across India.

Chapter 2
LITERATURE REVIEW

2.1 Use of Intelligent Monitoring Systems in Tourist Safety

2.1.1 Behavior Monitoring

A study by Zhang Wei (2019) explains how monitoring human movement patterns can help identify unusual behavior in public places. The study shows that changes such as irregular movement or sudden stops can indicate possible safety risks.Similarly, Kumar Ramesh (2021) focused on analyzing tourist movement using mobile data. The research found that tracking behavior patterns can help in identifying when a tourist is lost or entering unfamiliar areas.

2.1.2 Risk Detection

According to Li Chen (2020), analyzing location and movement data can help detect unsafe situations. The study highlights that identifying risks at an early stage can help in preventing serious incidents.

Another study by Singh Arjun (2022) shows that early risk detection plays an important role in improving safety. The research emphasizes that recognizing risky situations quickly allows faster action to be taken.

2.1.3 Alert Generation

Research by Ahmed Khan (2018) highlights the importance of automatic alert systems in safety applications. The study shows that alerts generated by the system can help reduce delays in communication.In addition, Priya Sharma (2021) explains that real-time alerts improve communication between users and authorities, making it easier to respond quickly during emergencies.

2.1.4 Real-Time Tracking

A study by John Peterson (2019) explains how real-time tracking helps in identifying the exact location of individuals during emergencies. This makes it easier for authorities to provide quick assistance.

Similarly, Anil Verma (2022) showed that continuous tracking improves response time and helps in better coordination during critical situations.

2.2 Integration of Location-Based Technologies and Safety Systems

Location-based technologies (LBT) play an important role in improving safety by allowing real-time tracking of user movements. These systems use GPS and mobile networks to find the exact location of tourists, which is very helpful when they are in unfamiliar places. One commonly used method is Geofencing, where virtual boundaries are created around safe and unsafe areas. When a tourist enters or leaves these areas, alerts are sent, helping them stay aware and avoid risky locations.

Many studies show that combining LBT with safety systems improves response time and safety. However, most systems work separately and lack proper integration, reducing their effectiveness. A connected system that combines tracking, alerts, and communication is needed for better tourist safety.

2.3 Limitations, Inconsistencies, Gaps, and Contradictions

2.3.1 Lack of System Integration

Most existing solutions focus on single features like tracking or alerts, but they are not combined into one unified system. This lack of integration reduces efficiency and makes it difficult to handle situations effectively.

2.3.2 Dependence on Manual Input

- Requires users to trigger alerts manually during emergencies
- May fail when the user is unable to respond or act quickly
- Increases delay in communication and response time

2.3.3 Security and Response Limitations

There are concerns related to data security, delayed response, and poor coordination between services. These issues reduce the overall effectiveness of existing safety systems.

2.4 Recommended Modifications

To overcome the limitations of existing systems, there is a need for a more connected and efficient solution. A system that brings together monitoring, tracking, identity management, and emergency response can improve safety and provide faster support. When all these features are integrated into a single platform, it becomes easier to manage information and handle situations quickly during emergencies.

2.4.1 Development of Integrated System

- System integration
- Real-time operation
- Better coordination

These features help in combining all parts of the system into one platform, making it easier to respond quickly and work more efficiently.

2.4.2 Improved Security and Access System

The system implements Firebase Authentication, Role-Based Access Control (RBAC), JWT token management, and Blockchain Digital ID verification to ensure secure, authorized access for tourists, police officers, and administrators across all dashboards.

2.5 Summary of Literature Review

Table 2.1: Comparative Analysis of Existing Tourist Safety Systems and Technologies

Author(s)	Year	Focus Area / Concept	Technology Used	Limitations / Gaps
Sharma et al.	2023	Tourist safety application	Mobile App, GPS	Manual alert activation
Li & Chen	2022	Real-time location tracking	GPS, IoT	No automatic risk detection
Kumar et al.	2024	Zone-based safety alerts	Geofencing, LBS	Not integrated with response systems
Ahmed Khan et al.	2021	Emergency communication system	Mobile Network	Delay in response
Ramesh Gupta et al.	2022	Digital identity system	Cloud, Database	Security and privacy issues
Priya Sharma et al.	2023	QR-based identity access	QR Code, Mobile App	Risk of duplication
Verma & Singh	2024	Device-based monitoring system	IoT, Sensors	High cost

Chapter 3
METHODOLOGY

3.1 Data Collection and Location Monitoring

The system collects real-time location data of tourists using GPS, which helps in continuously tracking their movement. This makes it easier to understand where a tourist is at any moment and whether they are moving safely. By observing location patterns, the system can identify if someone is entering an unfamiliar or potentially risky area, allowing timely support when needed.

Along with location tracking, the system also uses Geofencing to define safe and restricted zones for better monitoring. It provides QR code-based identity access, which helps in quick verification during emergencies. The system also includes an SOS feature that allows tourists to send instant alerts to authorities when they are in danger. By combining GPS tracking, geo-fencing, QR code access, and emergency alerts, the system ensures better safety and faster response in critical situations.

- Continuous tracking of tourist location in real time
- Monitoring movement between safe and unsafe zones
- Identifying unusual movement patterns
- Sending alerts during risky situations

3.2 System Development Model (V-Model)

The project follows the V-Model, which gives a clear and step-by-step way of developing the system. In this approach, each stage is planned carefully and checked before moving to the next step. It helps in identifying mistakes early and ensures that the system is built in an organized manner. By following this model, the development process becomes easier to manage and more reliable. It also makes sure that every part of the system is properly tested before it is used.

- Requirement Analysis – Understanding tourist safety needs

- System Design – Designing tracking, alert, and identity modules
- Implementation – Developing system components
- Testing – Checking functionality and performance

This approach ensures that the system works properly, remains stable, and is ready for real-time use.

3.3 Existing and Comparable Systems

Various existing systems such as mobile safety applications and tracking systems have been studied to understand how safety is currently handled. These systems mainly offer features like location tracking, emergency alerts, and basic communication support. They help users share their location and reach out for help during difficult situations. Some applications also include simple SOS options that allow users to send alerts quickly. Overall, these systems provide a basic level of support for user safety.

However, most of these systems work separately and are not connected with other important features. They do not combine tracking, identity verification, and emergency response in one place. Because of this, they are less effective, especially during serious situations where quick action is needed. This shows the need for a more complete system that brings everything together. Based on this understanding, the proposed system is designed to provide better safety and faster response.

3.4 Mapping the System to V-Model Stages

The proposed system is developed in a simple and organized way by following the stages of the V-Model. Each stage focuses on a specific part of the system, starting from understanding the requirements to final deployment. This step-by-step process helps in building the system in a clear and structured manner. It also makes it easier to check and correct mistakes at each stage. Overall, this approach helps in improving the quality and reliability of the system.

- Requirement Stage – Identifying safety features and system needs
- Design Stage – Creating system architecture and flow
- Development Stage – Implementing tracking, geo-fencing, and identity modules

- Testing Stage – Verifying system accuracy and response
- Deployment Stage – Making the system available for users

This structured mapping helps the system work smoothly and ensures better performance.

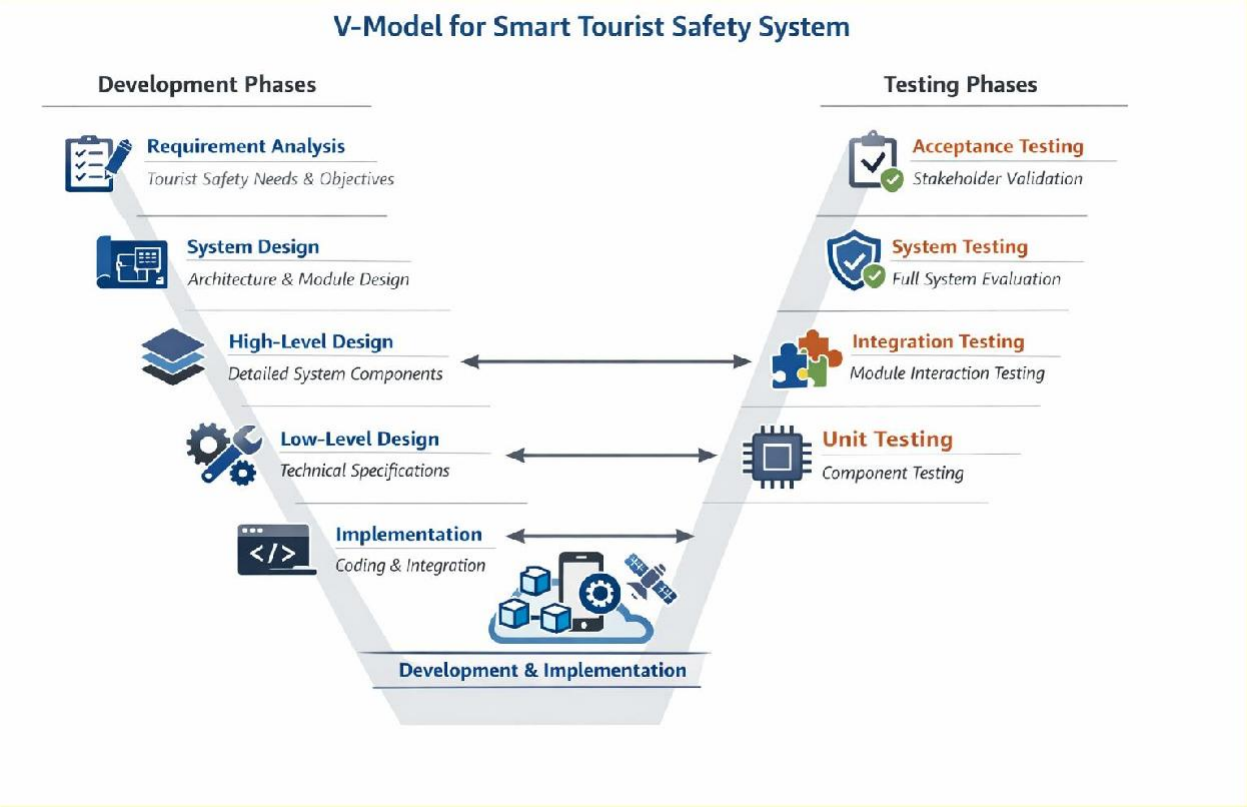

Fig 3.1: V-Model Methodology Applied to Tourist Safety Monitoring and Incident Response System

3.4 Mapping the System to V-Model Stages

The proposed system is developed in a simple and organized way by following the stages of the V-Model. Each stage focuses on a specific part of the system, starting from understanding the requirements to final deployment. This step-by-step process helps in building the system clearly and makes it easier to identify and fix issues. It also ensures that every part of the system is properly checked before moving to the next stage.

3.4.1 Requirement Stage

This stage focuses on identifying the needs of the system and understanding tourist safety requirements. It helps in defining what features are necessary for the system to work effectively.

3.4.2 Design Stage

In this stage, the overall structure of the system is planned, including how different components will work together. It focuses on designing modules such as location tracking, geo-fencing, and identity management. The data flow and communication between users and authorities are also defined. This stage ensures that the system is well-organized before development begins.

- System architecture planning
- Module structure design
- Data flow design

3.4.3 Development Stage

This stage involves building the system components such as tracking, geo-fencing, and identity features. It converts the planned design into a working system and ensures that all modules function properly.

3.4.4 Testing Stage

- Unit testing of modules
- System functionality testing
- Performance checking

3.4.5 Deployment Stage

In this final stage, the system is made available for users. It ensures that the system can be used in real-time situations and performs as expected.

Fig 3.2: Software Development Life Cycle (SDLC) Phases for Tourist Safety Monitoring System

3.5 Tools and Technologies Used

The system uses a combination of different technologies to ensure proper functioning, safety monitoring, and quick response during emergencies. Each technology plays an important role in making the system reliable and efficient.

- Geofencing – Used to define safe and restricted areas and monitor tourist movement
- GPS Tracking – Helps in tracking the real-time location of tourists
- SOS Emergency Feature – Allows users to send instant alerts during danger
- Incident Image Capture – Enables users to capture and send images to authorities
- QR Code – Used for quick identity access and verification
- Blockchain – Ensures secure storage of user identity data
- AI-based Alert System – Detects unusual situations and generates alerts automatically
- Internet / Network – Supports communication between users and authorities

The system is developed using React.js 18 for building interactive user interfaces and Vite as the build tool and development server. Tailwind CSS is used for responsive styling and React Router v6 handles client-side navigation. Firebase Authentication manages secure user login and registration using JWT tokens. Firebase Firestore serves as the real-time cloud database for storing incidents, alerts, and digital identity data. Google Maps API enables live location tracking and geo-fencing visualization. QR Code library generates blockchain-secured Digital ID codes. Recharts provides data visualization and React Hot Toast displays real-time notifications across all dashboards.

Chapter 4
PROJECT MANAGEMENT

4.1 Project Planning and Scheduling

Proper planning and scheduling play an important role in completing the project in an organized way. It helps in arranging tasks, setting timelines, and managing each stage effectively. By dividing the work into smaller steps, the team can focus on each task clearly. This also makes it easier to track progress and avoid confusion. Overall, it supports smooth and timely completion of the project.

The project was planned in different stages such as problem identification, requirement gathering, literature review, and system design. Each stage was given a fixed time period to maintain proper flow of work. This step-by-step approach helped in managing time efficiently and reducing delays. It also ensured that every part of the project was completed carefully. As a result, the entire development process became more structured and manageable.

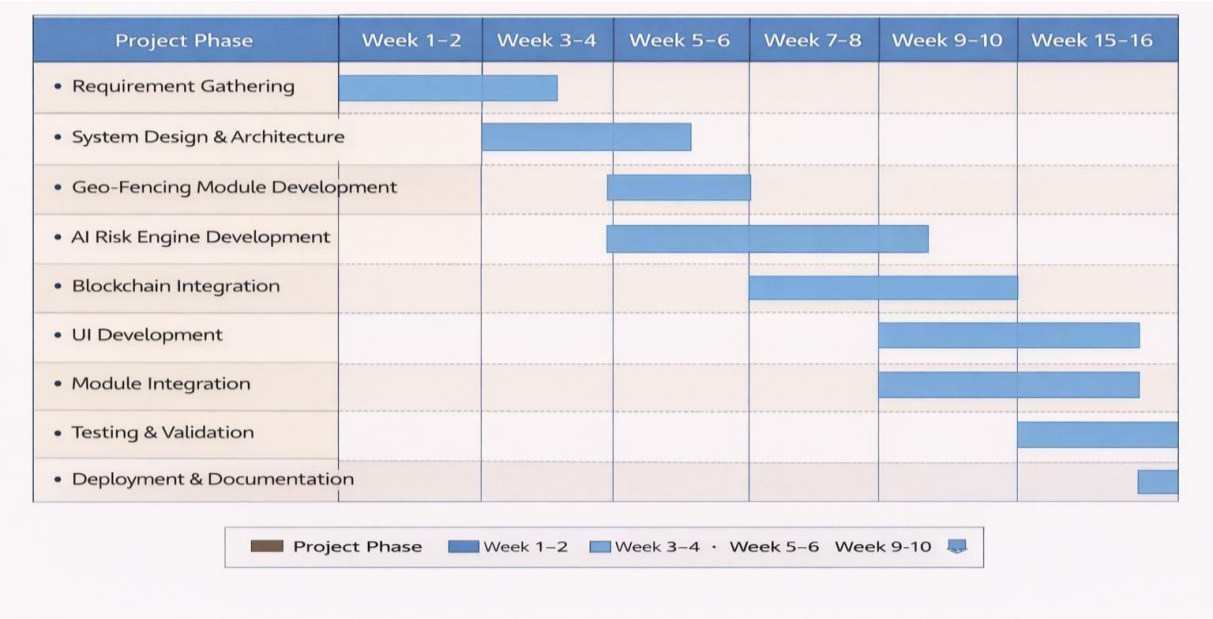

Fig4.1:Gantt Chart for Project Timeline

Table 4.1 Project Development Timeline and Milestones

Task	Start Date	End Date	Duration	Milestone / Output
Problem Identification	Week 1	Week 1	1 week	Problem Definition Finalized
Requirement Gathering	Week 1	Week 2	2 weeks	Requirement Specification
Literature Review	Week 1	Week 3	3 weeks	Literature Summary Completed
Selection of Methodology	Week 2	Week 3	2 weeks	Methodology Finalized
System Design Planning	Week 3	Week 4	2 weeks	Architecture Draft Prepared
Gantt Chart Preparation	Week 4	Week 4	1 week	Schedule Approved

The project planning and scheduling helped in clearly defining goals and managing time efficiently. It also made it easier to identify potential challenges at an early stage and take necessary actions. Overall, proper scheduling ensured that the project was completed in an organized and effective way.

4.2 Project Implementation Timeline

After completing the planning phase, the project moved into the implementation stage where the actual development work was carried out. This stage includes building different modules of the system, integrating them, and ensuring they work together properly. Tasks such as backend development, database integration, and system testing were performed step by step. Each activity was given a specific time period to maintain a proper workflow. This structured approach helped in completing the development process in an organized manner. It also ensured that each module was tested before combining them into the final system.

Table 4.2 Development Phase Timeline and Milestones

Task	Start Date	End Date	Duration	Milestone Output
Backend Module Development	Week 5	Week 8	4 weeks	Modules Completed
Frontend UI/UX Development	Week 6	Week 9	4 weeks	Interface Design Completed
Database Integration	Week 7	Week 9	3 weeks	Database Setup Completed

Module Integration	Week 10	Week 11	2 weeks	System Integration Completed
Testing (Unit + Integration)	Week 11	Week 12	2 weeks	Testing Completed
System Deployment	Week 12	Week 12	1 week	System Ready
Documentation & Review	Week 12	Week 13	1 week	Final Report Submitted

4.3 Risk Analysis

Risk analysis is important to understand the possible problems that may arise during the development and use of the system. These issues can affect how well the system works and how reliable it is in real situations. By identifying these risks early, it becomes easier to handle them and reduce their impact. Proper planning helps in avoiding major failures and ensures smooth functioning. This also improves the overall safety and effectiveness of the system.

4.3.1 Technical Risks:Problems related to system performance, errors, or integration may affect how the system works.

4.3.2 Data Security Risks:There is a chance of data being accessed without permission, especially sensitive user information.

4.3.3 Network and Connectivity Risks:Weak or unstable internet connection can delay alerts and communication during emergencies.

4.3.4 User Dependency Risks**:**The system may depend on user actions like sending alerts, which may not always happen in critical situations.

4.4 Software

The development of the *Smart Tourist Safety Monitoring & Incident Response System* requires the use of programming languages, data processing tools, and supporting libraries. These software components help in implementing features such as location tracking, alert generation, QR-based identity, and incident reporting. This section explains the main software tools used and their role in system development, testing, and deployment.

4.4.1 Programming Language

The main programming language used in this project is Python, which supports system development and data processing.

Python is used for:

- Location tracking and data handling
- Implementing geo-fencing logic
- Alert generation and SOS functionality
- QR code generation and processing
- Handling backend logic and system integration

Python provides strong library support, making it suitable for building efficient and scalable systems.

4.4.2 Libraries of Machine Learning and Data Science

NumPy

Used for numerical operations and handling data required for system processing.

Pandas

Used for data handling, cleaning, and organizing user and system data.

Scikit-learn

Used for basic data analysis and pattern recognition to support alert generation.

OpenCV

Used for capturing and processing incident images sent by users.

Matplot

Used for visualizing data such as movement patterns and system performance.

TensorFlow

Used for building models that help in detecting unusual situations and generating alerts.

QRCode

Used for generating and scanning QR codes for identity verification.

4.5 Backend & API Development

The backend is responsible for handling the main operations of the system and managing data flow between different components. It supports important features such as location tracking, alert generation, QR-based identity, and incident reporting. A well-structured backend ensures that the system works smoothly and responds quickly during real-time situations.

Flask/FastAPI

Flask is a lightweight and minimal Python web framework designed for simplicity and flexibility. It provides the essentials needed to build web applications and APIs without enforcing a strict structure, allowing developers to design the system their own way.

- Manage system functions and core processing
- Create APIs for communication between different modules

- Handle tracking logic and alert-related operations
- Coordinate and connect various components of the system

FireBase

It is commonly used to store user data, location tracking details, alerts, and media files while enabling seamless real-time synchronization across systems.

- User information and identity details
- Real-time location and tracking data
- SOS alerts and incident reports
- Images and system activity records

4.5.1 Frontend Development

The frontend is designed to make the system easy to use and understand. It provides a simple interface through which users can access different features without confusion. This helps users quickly view information and respond during emergencies.

HTML, CSS, JavaScript

- User location and tracking information
- Alerts and emergency notifications
- QR code for identity access
- Options to report incidents

React / Bootstrap

React is a JavaScript library used to build dynamic and interactive user interfaces through reusable components.Bootstrap is a CSS framework that helps create responsive, mobile-friendly designs quickly with pre-built styles and layouts.

4.6 Hardware

Even though the *Smart Tourist Safety Monitoring & Incident Response System* mainly focuses on software features, proper hardware support is necessary for smooth development and real-time operation. The system relies on devices that can handle location tracking, alert processing, and

communication effectively. This section describes the hardware components required to develop, test, and run the system.

4.6.1 Hardware Requirements for Development

Personal Computer/Laptop

Minimum Specification

- Processor: Intel i5 or AMD Ryzen 5
- RAM: 8 GB
- Storage: 256 GB SSD
- GPU: Integrated graphics
- Operating System: Windows 10 or Ubuntu

Recommended Specifications

For smoother performance and better development experience:

- Processor: Intel i7 or AMD Ryzen 7
- RAM: 16 GB or above
- Storage: 512 GB SSD
- GPU: Basic dedicated GPU (optional)
- Operating System: Windows or Linux

Purpose:
Used for coding, system development, backend setup, testing features like tracking, geo-fencing, QR code, and managing system data.

4.6.2 Virtual GPU/CPU Resources (Cloud Hardware)

Cloud resources are helpful in supporting the system by handling data, processing alerts, and managing backend operations. They make it easier to store information and ensure that the system

runs smoothly even when multiple users are active. Using cloud platforms also improves reliability and allows the system to handle real-time operations without delays.

Cloud Platforms (Optional but Recommended):

- Google Firebase / Google Cloud
- AWS (Amazon Web Services) /Microsoft Azure

Typical Cloud Specifications

Used to ensure reliable performance and scalability in cloud environments:

- vCPUs: 2–8 cores
- RAM: 8–32 GB
- Storage: Scalable cloud storage
- Network: High-speed internet connectivity

4.7 User Documentation

User documentation helps people understand how to use the system in a simple and clear way. It explains the basic features and guides users on how to access them without confusion. This makes it easier for anyone, even a new user, to use the system comfortably.

The documentation includes simple steps for using features like location tracking, sending SOS alerts, scanning QR codes, and reporting incidents. It also shows how the system responds during emergencies and how users can take action when needed.

Clear instructions make the system easier to use and reduce mistakes. It also helps users feel more confident while using the system, especially during important or emergency situation.

4.8 Training

Training helps users understand how to use the system properly and without confusion. It provides basic guidance on using features like location tracking, sending SOS alerts, scanning QR codes, and reporting incidents. Users are also shown how the system behaves during emergencies and how to respond quickly. This makes it easier for both tourists and authorities to use the system in real situations. Proper training reduces mistakes and improves confidence while using the system. It also ensures that the system can be used effectively when it is needed the most.

4.9 Testing

Testing is done to check whether the system works properly and all features run smoothly. It includes checking functions like location tracking, geo-fencing alerts, SOS alerts, QR code access, and incident reporting. Each feature is tested step by step and then checked together as a complete system. This helps in finding any issues and fixing them before the system is used. It also makes sure that the system responds quickly and correctly during emergencies. Proper testing improves the reliability of the system and makes it more dependable in real situations.

4.10 Go Live

The Go Live stage is the final step where the system is made available for real-time use. After completing development and testing, the system is deployed so that users and authorities can start using it. This stage ensures that all features are working properly in a real environment.

During this phase, the system is monitored to check its performance and identify any minor issues. Necessary adjustments are made to improve stability and ensure smooth operation. Once everything is functioning correctly, the system is fully ready to support users and provide safety assistance during real situations.

After deployment, continuous monitoring and feedback from users help in further improving the system. Any small issues or updates can be handled based on real-time usage. This helps in maintaining system performance and ensures that it remains reliable over time.

Chapter 5
ANALYSIS AND DESIGN

This section explains how the system requirements are identified and how the system is planned based on those needs. It focuses on understanding what users expect from the system and turning those needs into a clear and structured design. Proper analysis helps in building a system that works smoothly and supports safety features effectively. The design stage also makes sure that all parts of the system are properly arranged and connected. This helps in creating a system that is easy to manage and use.

5.1 Requirements

The requirements explain what the system should do and how it should work in real situations. They are identified based on the needs of tourists and authorities to ensure safety, quick response, and easy access to important information. The system includes features like location tracking, geo-fencing, SOS alerts, QR code-based identity, and incident reporting. These features help in monitoring tourist movement and providing support during emergencies.

Along with system features, the requirements also focus on performance and reliability. The system should run smoothly without delays and respond quickly during critical situations. It should keep user data safe and be simple to use so that anyone can access it without difficulty. Overall, these requirements help in building a system that is reliable, efficient, and suitable for real-world use.

Table 5.1: System Components and Functional Description

Component	Description / Function
Location Tracking Module	Uses GPS to track real-time location of tourists
Geo-Fencing Module	Defines safe and restricted zones and generates alerts
QR Code Module	Provides quick identity access and verification
SOS Alert Module	Sends emergency alerts to authorities instantly
Incident Reporting Module	Allows users to capture and send incident images
Backend Framework	Flask / FastAPI used for system logic and API handling
Database (Firebase)	Stores user data, location details, and alert information
Frontend Interface	HTML, CSS, JavaScript for user interaction and display
Alert & Notification System	Sends real-time alerts and updates to users and authorities
Security Module	Ensures safe handling of user data and identity information

5.2 Block Diagram

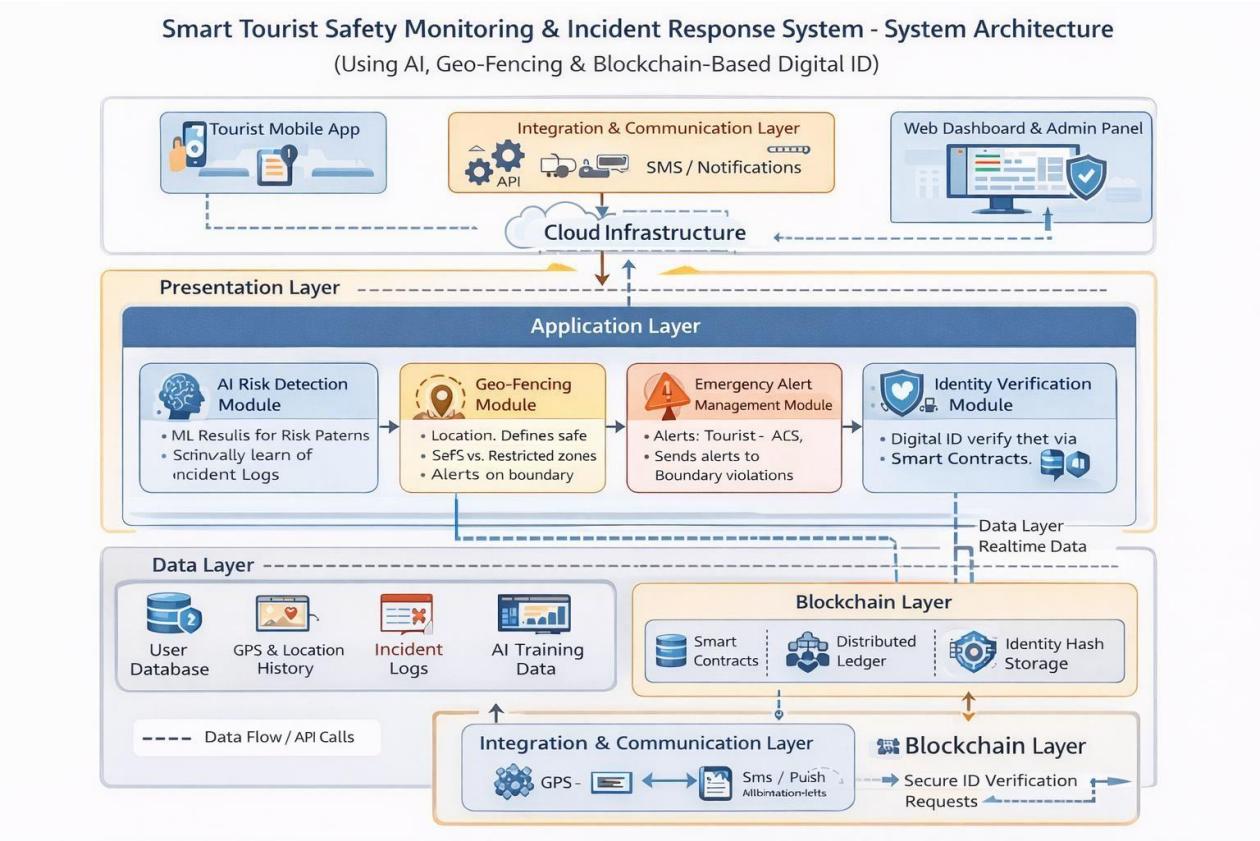

Fig 5.1 System Architecture of Smart Tourist Safety Monitoring and Incident Response System

The Smart Tourist Safety Monitoring and Incident Response System is structured into six distinct layers, each serving a specific role in the overall system architecture. At the top of the diagram lies the User Layer, which consists of three types of users — Tourists, Police Officers, and Administrators. Tourists access the system through a mobile or web application to track their location, send SOS alerts, and report incidents. Police Officers use a dedicated web dashboard to monitor tourist activities and respond to emergencies in real time. Administrators manage the entire system through an admin panel, overseeing geo-fence zones, user management, and system-wide analytics. All three user roles interact with the system through the Frontend Layer, which is built using React 18, Vite, and Tailwind CSS, providing a fast, responsive, and visually consistent interface across all devices.

The middle section of the block diagram represents the Core Modules and Services Layer, which forms the functional heart of the system. The Core Modules include GPS Tracking for real-time location monitoring using the useGeolocation hook, an SOS Alert module for instant emergency notifications, a Geo-Fencing module that automatically triggers alerts when tourists enter high-risk zones, an Incident Reporting module that allows tourists to submit detailed reports with auto-filled location data, and a Blockchain Digital ID module for secure and tamper-proof identity verification using QR codes. These modules are supported by the Services Layer, which includes Firebase Authentication for secure login via email or Google Sign-In with JWT token management, an Axios API Service for handling all REST API communication with request interceptors, and an Analytics Service powered by Recharts to generate visual trend reports and statistics for administrators

The bottom section of the diagram represents the Backend and Database Layer, which handles all data processing, storage, and retrieval operations. The REST API Server acts as the central processing unit, managing incident records, alert processing, user management, and geo-fence zone control. It communicates bidirectionally with three separate databases — the Firebase Database, which stores user profiles and authentication records; the Application Database, which maintains incident reports, SOS alerts, and geo-fence zone data; and the Blockchain ledger, which stores immutable digital identity records and system logs. This layered architecture ensures that the system is scalable, secure, and maintainable, with each layer having a clearly defined responsibility that contributes to the overall goal of ensuring tourist safety across India.

Table 5.2 Hardware and Software Components Used in the Smart Tourist Safety Monitoring System

Device	Purpose	Reason Selected
Laptop/PC	System development	Supports coding, testing, and system setup
Smartphone	User access and tracking	Easy to use for tourists in real-time
GPS Module	Location tracking	Provides accurate real-time position data
Cloud (Firebase)	Data storage and management	Enables secure and scalable data handling
Camera (Mobile)	Incident image capture	Helps in reporting real-time incidents
QR Code System	Identity verification	Quick and easy access to user information
Internet/Wi-Fi	Communication and data transfer	Supports real-time alerts and updates

5.3 System Flow Diagram

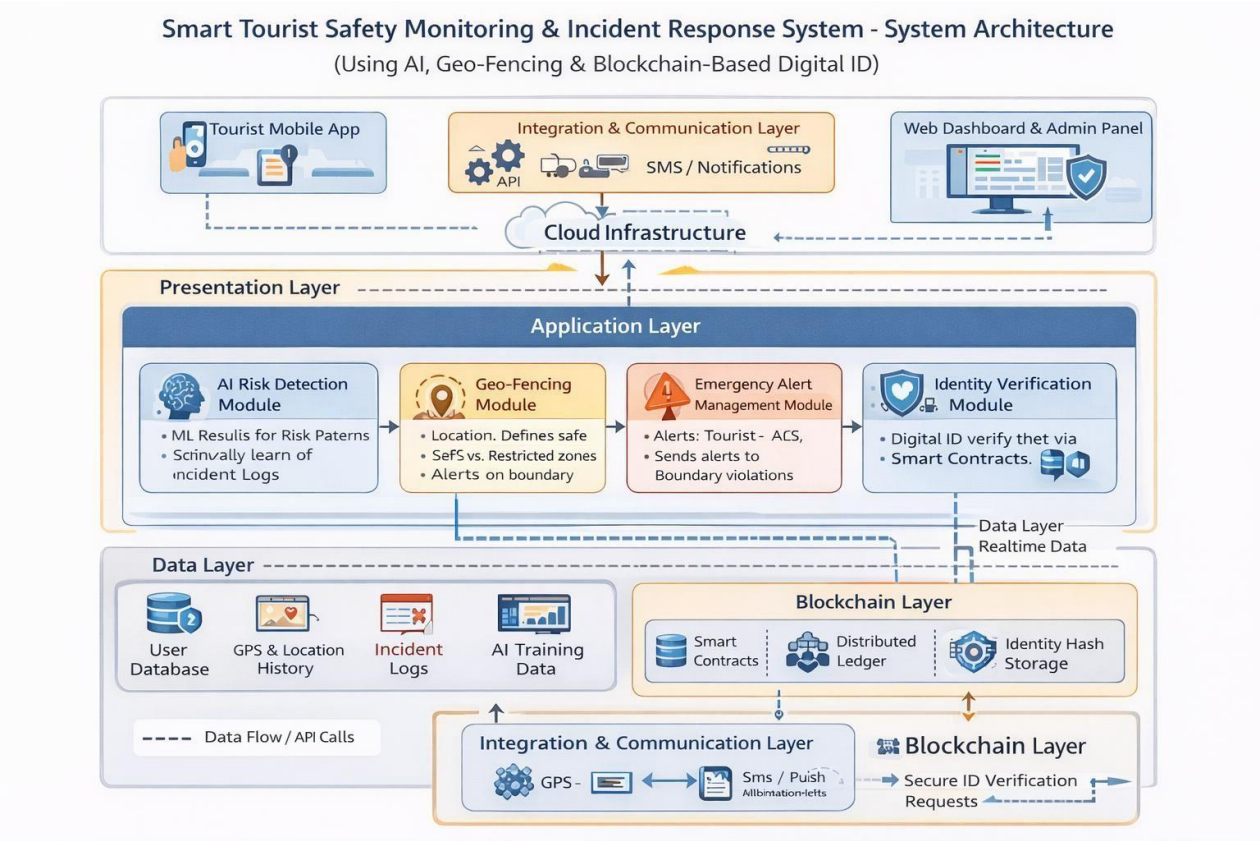

Fig 5.2 System Architecture of Smart Tourist Safety Monitoring Application

The system flow diagram shows how the tourist safety system works in a simple step-by-step manner. It starts with the tourist registering in the system and receiving a QR code for identification. After registration, the system tracks the tourist's location using GPS and keeps monitoring their movement. If the tourist enters a restricted area or sends an SOS alert, the system immediately sends a notification. The admin checks the alert and passes the information to the police, who then take action to help the tourist. If no issue is detected, the system continues to monitor the location. This flow helps in ensuring safety, quick alerts, and timely response during emergencies.

5.4 Choosing Devices

The selection of devices for this project is based on what is needed for smooth working and quick response in real situations. The system mainly uses simple and easily available devices that can support tracking, communication, and safety features. This makes the system practical and easy to use without depending on complex hardware.

Smartphones are the main devices used, as they help in tracking location, sending SOS alerts, capturing incident images, and accessing QR-based identity. GPS is used to provide accurate location details, while internet connectivity helps in sending data and alerts without delay. Cloud services are used to store and manage information safely.Overall, the devices are chosen to be simple, reliable, and easy to access. This helps in making the system effective and suitable for real-world use, especially during emergency situations.

5.5 Designing Units

The system is divided into different units, where each unit performs a specific task to ensure smooth working. It includes units for collecting user details and location data, processing information like tracking and geo-fencing, generating alerts such as SOS notifications, and handling responses through admin and police. There is also a storage unit that manages all data like user information and incident records. This division makes the system easier to manage and understand. It also allows changes or updates to be made to one part without affecting the whole system, making it more flexible and efficient for real-world use.

5.6 Standards

The project is developed by following proper practices to make sure the system works smoothly and is easy to use. These guidelines help in building a system that is clear, reliable, and meets user needs. During development, care is taken to define all features properly and organize them in a simple and structured way.Location data and other information are handled in a structured format so that they can be easily managed and shared within the system. Clear documentation is also maintained to explain how the system works and what features it provides. Following these practices makes the system easier to maintain and helps in making future updates or improvements without difficulty.

5.7 Mapping with IoTWF Reference Model Layers

The system can be understood using IoT layers such as device, communication, data processing, and application layers. In this project, the smartphone acts as the device that collects location and user data. This data is sent through network communication to the backend, where it is processed and stored. The final information is then used in the application layer to generate alerts and responses. This layered approach helps in organizing the system and makes it easier to manage and improve.

5.8 Domain Model Specification

The domain model defines the main elements of the system such as user, location, alerts, QR identity, and incident reports. It shows how these elements are connected and how data flows between them. This helps in understanding how information is stored, processed, and used in the system. It also makes the system flexible, allowing new features to be added easily in the future.

5.9 Communication Model

The communication model explains how information moves between different parts of the system. User data and location details are sent to the backend through APIs, where they are processed. Alerts and responses are then sent back to users, admin, and police. This ensures smooth and continuous communication within the system. It also supports quick data transfer during emergency situations.

5.10 IoT Deployment Level

The system follows a cloud-based deployment approach where data is collected from user devices and processed in the backend. It supports real-time tracking, alert generation, and data storage. This setup makes the system flexible and scalable, allowing it to handle more users in the future. It also makes it easier to upgrade the system with additional features if needed.

5.11 Functional View

The functional view explains what the system does from the user's perspective. It includes features like registration, QR code generation, location tracking, geo-fencing alerts, SOS alerts, and incident reporting. It also includes storing user data and alert history. This view helps in understanding the purpose and working of the system clearly.

5.12 Mapping IoT Deployment Level with Functional View

This section connects the system functions with the overall architecture. The device layer collects data, the backend processes it, and the application layer shows results and alerts to users. Each function is linked to a specific part of the system. This helps in maintaining a clear structure and ensures proper implementation.

5.13 Operational View

The operational view describes how the system works in real-time. It shows how data is collected, processed, and used to generate alerts and responses. It also includes aspects like data validation, error handling, and system performance. This ensures that the system works reliably and responds quickly during emergencies.

5.14 Other Design

Other design aspects focus on making the system simple and easy to use. The interface is designed to be clear, with easy navigation and quick access to features like SOS alerts and QR code. The system is also designed in a modular way so that different parts can be updated without affecting the whole system. In the future, additional features like maps, mobile apps, or advanced alerts can be added to improve usability and performance.

Chapter 6
HARDWARE, SOFTWARE AND SIMULATION

6.1 Hardware

The hardware setup of the *Smart Tourist Safety Monitoring & Incident Response System* is kept simple and easy to use. It is divided into different parts, where each part handles a specific task such as tracking location, sending alerts, and reporting incidents. These parts work together to make the system function properly. This kind of setup also makes it easier to manage and test the system.

The system mainly uses smartphones and basic computing devices for its operation. Smartphones are used for tracking location, sending SOS alerts, capturing incident images, and accessing QR-based identity. GPS helps in providing accurate location details, and internet connectivity allows smooth communication between users, admin, and police. The use of simple and easily available devices makes the system practical for real-world use.

Overall, the hardware design focuses on reliability and ease of use. It supports continuous monitoring and quick response during emergencies. The system is also flexible, so it can be improved or expanded in the future without much difficulty.

6.2 Software and Simulation

The software and simulation phase is important to check how the system works before it is used in real situations. It helps in testing different features and making sure everything runs smoothly. By using suitable software tools, the system is developed and verified step by step. This process also helps in finding any issues early and improving the overall performance.

Simulation is done by creating real-life situations such as tracking movement, entering restricted areas, or sending SOS alerts. The system's response is then observed to see how quickly and

correctly it reacts. This helps in making sure the system works properly and provides reliable support when needed.

6.2.1 Programming Language

JavaScript (React.js) JavaScript is used as the main programming language for developing the system. React.js framework is used to build interactive and dynamic user interfaces. It supports features like real-time tracking, geo-fencing alerts, incident reporting, and role-based dashboard operations.

6.2.2 Supporting Libraries

React Router v6 Used for managing client-side navigation and routing between different pages and dashboards within the application. Tailwind CSS Used for designing and styling the user interface with a responsive and modern layout across all devices. QR Code Library (qrcode.react) Used for generating and displaying QR codes for tourist Digital ID verification at checkpoints and security locations. Recharts Used for building interactive charts and visual analytics in the administrator dashboard. React Hot Toast Used for displaying real-time success, error, and notification messages across the application. Heroicons Used for providing clean and consistent SVG icons throughout the user interface. Axios Used for handling HTTP requests and managing communication between the frontend and backend services.

6.2.3 Processing and System Support

Firebase Authentication Used for managing secure user registration, login, and session management using email/password and Google Sign-In. Firebase Firestore Used for storing and retrieving real-time data including user profiles, incident reports, SOS alerts, digital IDs, and geo-fence information. Firebase Storage Used for securely storing incident images uploaded by tourists during incident reporting. Google Maps API Used for displaying real-time location tracking, geo-fence zones, tourist markers, and interactive maps across all dashboards. API Integration Used

for sending and receiving data between tourists, police officers, and administrators in real time, ensuring seamless communication across all system components. Vite Used as the build tool and development server for fast compilation and optimized production deployment of the application.

6.2.4 Simulation Results

- Accurate real-time location tracking
- Detection of entry into restricted areas
- Quick alert generation during risky situations
- Proper working of SOS emergency alerts
- Successful incident image capture and reporting
- Reliable system response in different scenarios

6.3 Software Development Tools

The development of the *Smart Tourist Safety Monitoring & Incident Response System* uses a combination of tools for coding, data handling, and system management. Python is used as the main programming language because it is simple and supports various system features. Visual Studio Code is used for writing and organizing the code, while libraries like NumPy and Pandas help in handling data. Firebase is used to store user information, location details, and alerts, and tools like Git and GitHub help in managing code updates and version control. All these tools work together to make the development process smooth and well-organized.

6.4 Software Code

6.4.1 Firebase Configuration

```
const firebaseConfig = {

  authDomain: "smart-tourist-safety-32136.firebaseapp.com",

  projectId: "smart-tourist-safety-32136",
```

```
  storageBucket: "smart-tourist-safety-32136.firebasestorage.app",
  messagingSenderId: "217905774812",
  appId: "1:217905774812:web:c0a07b4ac5e27fbf4fb419"
};
const app = initializeApp(firebaseConfig);
export const auth = getAuth(app);
export const db = getFirestore(app);
```

6.4.2 User Authentication

```
// Register
export const registerWithEmail = (email, password) =>
  createUserWithEmailAndPassword(auth, email, password);


// Login
export const loginWithEmail = (email, password) =>
  signInWithEmailAndPassword(auth, email, password);


// Google Login
export const loginWithGoogle = () =>
  signInWithPopup(auth, googleProvider);
```

```
// Logout

export const logout = () => signOut(auth);
```

6.4.3 Protected Route

```
const ProtectedRoute = ({ children, allowedRoles }) => {

  const { user, role } = useAuth();

  if (!user) return <Navigate to="/login" replace />;

  if (!allowedRoles.includes(role))

    return <Navigate to="/" replace />;

  return children;

};
```

6.4.4 Create Incident (Firestore)

```
export const createIncident = async (incidentData) => {

  const docRef = await addDoc(collection(db, 'incidents'), {

    ...incidentData,

    timestamp: serverTimestamp(),

    status: 'Active',

    severity: 'Medium'

  });
```

```
  return docRef.id;
};
```

6.4.5 SOS Alert

```
export const createAlert = async (alertData) => {
  const docRef = await addDoc(collection(db, 'alerts'), {
    ...alertData,
    timestamp: serverTimestamp(),
    status: 'Active'
  });
  return docRef.id;
};
```

6.4.6 Digital ID with QR Code

```
const qrData = JSON.stringify({
  id: mockDigitalID.idNumber,
  name: digitalIDData.fullName,
  nationality: digitalIDData.nationality,
  digiLocker: digitalIDData.digiLockerId,
  emergency: digitalIDData.emergencyContact,
  verified: true
```

```
});


<QRCodeSVG value={qrData} size={128} level="H" />
```

6.4.7 Geo-Fencing Zones

```
const geoZones = [
  { name: 'India Gate', type: 'safe',
    color: '#22c55e', radius: 500 },
  { name: 'Connaught Place', type: 'hub',
    color: '#3b82f6', radius: 600 },
  { name: 'Old Delhi', type: 'restricted',
    color: '#ef4444', radius: 700 },
];


new window.google.maps.Circle({
  strokeColor: zone.color,
  fillColor: zone.color,
  fillOpacity: 0.25,
  map: googleMapRef.current,
  center: { lat: zone.lat, lng: zone.lng },
```

```
  radius: zone.radius,

});
```

6.5 Simulation

Simulation is used to check how the system works in real-life situations before it is actually used. It helps in making sure that features like tracking, geo-fencing, alerts, and incident reporting are working properly. By testing different situations, it becomes easier to see how the system responds and to ensure that it performs smoothly and reliably.

6.5.1 System Testing

The system is tested by checking different features such as location tracking, QR code access, and SOS alerts. Each feature is used step by step to see whether it is working correctly. The system is also observed to ensure that alerts are generated properly and sent to the admin and police without delay.

Any issues found during testing are identified and fixed to improve system performance.

6.5.2 Performance Evaluation

The system performance is checked based on how quickly it responds, how accurately it tracks location, and how reliably it sends alerts. It also looks at how fast and correctly notifications are delivered during emergency situations.

This helps in making sure the system works smoothly and responds properly in real-time situations.

6.5.3 Scenario Analysis

Different situations are tested, such as a tourist entering a restricted area, sending an SOS alert, or reporting an incident. These tests help in understanding how the system reacts in different conditions and whether it responds correctly.This ensures the system can handle real situations and provide quick support when needed.

The system was tested under various real-world scenarios including tourist SOS alerts, incident reporting, geo-fence zone detection, Digital ID creation, police alert management, admin monitoring, government portal access, and unauthorized access attempts. All scenarios performed successfully, confirming the system operates correctly and reliably under different situations.

When a tourist enters a restricted geo-fence zone, the system triggers an automatic alert. Police officers receive real-time SOS notifications and respond instantly. Administrators monitor live tourist locations and manage safety zones effectively. All tested scenarios including incident reporting, Digital ID verification, and emergency response produced accurate and expected results.

The Smart Tourist Safety Monitoring and Incident Response System was tested under various real-world scenarios to evaluate its performance, reliability, and accuracy across all user roles. In the first scenario, a tourist in an emergency situation clicks the SOS button on the dashboard. The system instantly captures the GPS location, stores the alert in Firebase Firestore, and notifies the police dashboard in real time. The police officer accepts the alert and dispatches a response unit immediately. In the second scenario, a tourist reports a theft incident by filling the incident report form with description, location, and photo evidence. The system successfully stores the report in Firestore and displays it on both police and admin dashboards instantly. In the third scenario, a tourist approaches a restricted geo-fence zone. The system automatically triggers a safety alert, warning the tourist to avoid the high-risk area while simultaneously notifying the police dashboard. In the fourth scenario, an unauthenticated user attempts to access the police dashboard directly. The protected route system immediately detects unauthorized access and redirects the user to the login page, successfully preventing any security breach. All scenarios produced accurate and expected results, confirming the system performs reliably under real-world conditions.

Chapter 7
EVALUATION AND RESULTS

7.1 Test Points

The test points are used to check how well the system works in different situations. They help in making sure that all features are working properly and the system responds correctly when needed. Each part of the system is tested to ensure smooth operation and reliable performance in real-time use.

7.1.1 Input

Data Input Validation ensures that the system correctly accepts, processes, and stores user-entered data. The following validations are performed across the system.

7.1.2 Tracking Accuracy

Ensures that the system shows the correct real-time location of the tourist.

7.1.3 Alert Generation Accuracy

Checks whether alerts are sent correctly when a tourist enters a restricted area or uses the SOS feature.

7.1.4 System Performance

Verifies how quickly the system responds and sends alerts during different situations.

7.1.5 Real-time Response Check

Checks the speed at which alerts reach the admin and police.

7.1.6 Error Handling

Ensures the system handles wrong or missing inputs properly and shows correct messages.

7.1.7 Output Consistency

Checks whether the system gives the same result when the same input is given multiple times.

7.2 Test Plan

The test plan explains how the system will be checked to make sure everything is working properly. It includes the methods used to test different features like tracking, alerts, QR code, and incident reporting. This helps in understanding whether the system performs well in different situations.

Goal: To make sure the system works correctly, responds quickly, and provides reliable support through features like location tracking, geo-fencing, SOS alerts, QR code access, and incident reporting in real-time conditions.

7.3 : Test Cases

Table 7.1 Test Cases for Tourist Safety Monitoring and Incident Response System

Test Case ID	Description	Input	Expected Output
TC01	User input validation	Invalid or incomplete user details	Error message displayed
TC02	QR Code generation	User registration details	QR code generated successfully

TC03	Location tracking	GPS enabled device	Accurate real-time location displayed
TC04	Geo-fencing alert	Entry into restricted area	Alert generated and sent
TC05	SOS alert functionality	User triggers SOS	Immediate alert sent to admin and police
TC06	Incident reporting	Image capture and report submission	Incident details sent successfully
TC07	System response	Alert generated	Quick notification to admin and police
TC08	Error handling	Missing or incorrect input	Proper error message displayed

Test Results

Table 7.2 Test Execution Results for Tourist Safety Monitoring and Incident Response System

Test Case ID	Description	Expected Result	Actual Result

TC01	User input validation	Show error for invalid input	Error message displayed correctly
TC02	QR code generation	QR code generated successfully	QR code generated without error
TC03	Location tracking	Accurate real-time location	Location tracked correctly in real-time
TC04	Geo-fencing alert	Alert when entering restricted area	Alert generated and sent successfully
TC05	SOS alert	Immediate alert to admin and police	Alerts delivered quickly and correctly
TC06	Incident reporting	Image and report sent successfully	Incident data received by system
TC07	System response	Fast notification during emergencies	Response time within 1–2 seconds

TC08	User interface	Accept input and display results	Interface works correctly and displays data

Observation

- The system works accurately and gives reliable results in different situations
- Location tracking and alert features run smoothly without issues
- The system responds quickly, which helps during real-time use
- No major problems were noticed while testing the system

7.4 Insights

The observations from testing and evaluation show how effectively the system works and where it can be improved further.

7.4.1 System Performance Insight

The system performs consistently and responds quickly in different situations. Features like tracking, geo-fencing, and alerts work properly, making the system reliable for real-time use.

7.4.2 Data Insights

Location data plays a key role in the system, especially in tracking movement and identifying restricted areas. Combining location with geo-fencing improves the accuracy of alerts and overall safety monitoring.

7.4.3 System Reliability Insight

The system handles different situations smoothly, even when there are minor issues like weak signals or delayed inputs. It gives stable results and works consistently during repeated use.

7.4.4 Usability Insight

The system is simple to use, and users can easily access features like SOS alerts, QR code, and incident reporting. It does not require much training, which makes it suitable for all users.

7.4.5 Real-World Application Insight

The system can help improve tourist safety by providing quick alerts and support during emergencies. It reduces response time and helps authorities take faster action when needed.

The Smart Tourist Safety Monitoring and Incident Response System has significant real-world applications in India's growing tourism industry. With millions of domestic and international tourists visiting popular destinations every year, ensuring their safety has become a top priority for government authorities and law enforcement agencies. In real-world deployment, this system can be integrated with state and central government tourism departments to provide comprehensive safety coverage across major tourist destinations. Police control rooms can monitor all active tourists on a live map, respond to SOS alerts instantly, and manage incident reports efficiently without manual intervention. The blockchain-secured Digital ID system eliminates the need for physical document verification at checkpoints. Authorities can simply scan the tourist's QR code to instantly verify their identity, emergency contacts, and travel details, significantly reducing verification time during emergencies. The geo-fencing feature can be practically implemented across high-risk areas, crowded markets, and tourist hotspots in cities like Delhi, Mumbai, Jaipur, and Bengaluru. Automatic alerts notify both tourists and police when safety boundaries are crossed. Furthermore, direct integration with Karnataka Tourism and Government of India Tourism portals provides tourists with authentic travel information and government advisories, making this system a complete and reliable real-world tourist safety solution.

This system can be practically deployed across major Indian tourist destinations like Delhi, Mumbai, Jaipur, and Bengaluru. Government authorities, police departments, and tourism boards can adopt this platform to monitor tourist safety, respond to emergencies instantly, and provide a secure and confident travel experience for all visitors.

Chapter 8
SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY

The development of the *Smart Tourist Safety Monitoring & Incident Response System* is not only about building a technical solution but also about using it responsibly. Since the system deals with user location, identity, and emergency data, it is important to ensure privacy and proper usage. The system should protect user information and avoid misuse of data. It also needs to follow basic legal rules related to data handling and safety. From a social point of view, the system helps in improving tourist safety and building confidence. Overall, considering these aspects makes the system safe, reliable, and suitable for real-world use.

8.1 Social Aspects

Social aspects focus on how the system helps people and improves safety in real situations. The main aim is to make tourists feel secure while traveling and provide support when needed.

• User Safety: The system helps tourists stay safe by providing features like tracking and emergency alerts.

• Privacy: User information is handled carefully to keep personal data safe.

• Trust: Reliable alerts and correct information help users feel confident while using the system.

• Support for Authorities: The system helps police and admin respond quickly during emergencies.

• Community Benefit**:** Better tourist safety also helps create a safer environment for everyone.

8.2 Legal Aspects

Legal aspects focus on making sure the system is used properly and follows basic rules related to data and safety. Since the system uses location and personal details, it is important to handle this information carefully.

• Data Protection: User information is kept safe and protected from unauthorized access.

• User Permission: Users are informed about data usage and their consent is taken before using the system.

• Data Safety: Proper steps are taken to store and manage data securely.

• Proper Usage: The system is used only for safety purposes and not for any misuse.

• Compliance: The system follows general guidelines to ensure safe and responsible use.

8.3 Ethical Aspects

Ethical aspects focus on using the system in a responsible and fair way. Since the system deals with personal information and location data, it is important to respect user privacy and avoid misuse.

• User Privacy: The system respects user privacy and avoids unnecessary tracking.

• Proper Use: The system is used only for safety and emergency support.

• Transparency: Users are clearly informed about how their data is used.

• Fairness: The system treats all users equally and provides the same level of support.

• Accuracy: The system provides correct information to avoid confusion during emergencies.

8.4 Sustainability Aspects

Sustainability focuses on making sure the system is efficient and does not create unnecessary impact while being used over time. The system is designed to use available resources carefully and support long-term usage without issues.

• Efficient Resource Use: The system uses minimal hardware and relies on common devices like smartphones, reducing extra resource usage.

• Energy Usage: Cloud services and devices are used in a way that avoids unnecessary energy consumption.

• Digital Approach: Since the system is mostly digital, it reduces the need for physical materials and paperwork.

• Scalability: The system can be expanded easily without requiring major changes or additional resources.

• Long-term Use: The design supports future updates and improvements, making the system usable for a longer time.

8.5 Safety Aspects

Safety aspects focus on protecting users and ensuring that the system works properly during real situations. The system is designed to support tourists and provide quick help during emergencies.

• User Safety: The system helps tourists stay safe by providing features like SOS alerts, location tracking, and quick support.

• Emergency Response: Alerts are sent quickly to admin and police so that help can reach the user without delay.

• Data Safety: User information is handled carefully to avoid misuse or unauthorized access.

• System Reliability: The system is designed to work properly without errors, especially during critical situations.

• Safe Usage**:** The system is simple to use, reducing the chances of mistakes during emergencies.

Chapter 9
SUMMARY

This project presents a Smart Tourist Safety Monitoring and Incident Response System designed to improve the safety of tourists in real-life situations by using simple and practical technologies. The system focuses on providing quick support and reducing risks through features like real-time location tracking, geo-fencing, SOS alerts, QR-based identity, and incident reporting. Tourists can register in the system and receive a QR code, which helps in easy identification when required. The system continuously tracks the tourist's location using GPS and monitors movement to detect whether the user is in a safe or restricted area. If a tourist enters an unsafe zone or triggers an SOS alert, the system immediately sends notifications to the admin and police. This helps authorities respond quickly and provide necessary assistance without delay. The system also allows users to capture and send incident images, which improves communication and helps in better decision-making during emergencies. The project is developed using commonly available devices like smartphones, making it easy to use and practical for real-world implementation. It is supported by software tools such as Python and Firebase, which help in managing data and system operations efficiently. Proper testing and simulation were carried out to check the performance of the system in different situations. The results show that the system works smoothly, provides accurate tracking, and responds quickly in emergency cases. The design of the system is kept simple so that users can easily understand and use it without confusion. The project also considers important aspects such as user safety, data privacy, and proper usage of information. These factors help in building trust among users and ensure that the system is used responsibly. Overall, the system provides a useful and effective solution for improving tourist safety and supports faster response during emergencies. It also creates a safer environment for travelers and helps authorities manage situations better. In the future, the system can be improved by adding more advanced features and expanding its usage to a larger scale.

9.1 Summary of Approach

The system is designed to improve tourist safety by bringing different features together into one simple and connected platform. It starts with user registration, where each tourist receives a QR

code for easy identification. After registration, the system tracks the tourist's location using GPS and monitors movement through geo-fencing to identify safe and restricted areas. This helps in noticing any unusual or risky situations. If a tourist enters an unsafe area or faces a problem, the system can generate alerts or allow the user to send an SOS message. These alerts are quickly shared with the admin and police so that help can be provided without delay. The system also allows users to capture and send incident images, which helps authorities understand the situation better. The backend manages all the data and handles communication between different parts of the system. The admin keeps track of activities and alerts, while the police respond to emergencies based on the information received. The system uses simple devices like smartphones, making it easy to use in real situations. Cloud services are used to store data and support communication. The design is modular, so different parts of the system can be updated or improved without affecting the whole system. Testing is done to check how well features like tracking and alerts work in different situations. The results show that the system works smoothly and responds quickly when needed. The approach focuses on simplicity, quick response, and ease of use. It also considers user safety, data privacy, and proper usage. This makes the system practical and useful for real-world applications and helps create a safer environment for tourists.

9.2 Future Improvements

The system can be improved further by adding new features and enhancing its current capabilities. Future updates can focus on making the system more advanced, user-friendly, and effective in real-time situations.

Add map-based tracking using GIS to show tourist location and unsafe zones clearly

Improve alert system using advanced models for better detection of risky situations

Include real-time data collection from devices for more accurate monitoring

Expand the system to support larger areas and more users

Develop a mobile application for easier access and better user experience

Add multi-language support to make the system usable for different users

Include user feedback to improve system performance over time

Show alert levels or risk levels for better understanding of situations

9.3 Summary of Results

The results show that the system works effectively and performs well in different situations. Features like location tracking, geo-fencing, SOS alerts, and incident reporting function correctly and provide timely responses. The system is able to detect unsafe conditions and send alerts quickly to the admin and police, helping in faster action during emergencies.Testing confirms that the system is reliable, easy to use, and suitable for real-time applications. The response time is quick, and the overall performance remains stable even when tested multiple times. The system also handles different scenarios smoothly without major issues. Overall, the results indicate that the system is practical, dependable, and useful for improving tourist safety.

System Performance:The system works smoothly and all features operate as expected in different situations.

Tracking Accuracy:Location tracking is accurate and helps in monitoring tourist movement in real time.

Alert Response:The system quickly sends alerts to admin and police during emergencies.

Reliability:The system performs consistently without major issues during repeated testing.

Ease of Use:The system is simple to use and users can access features without difficulty.

Overall Outcome:The results show that the system is reliable, efficient, and useful for improving tourist safety.

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BASE PAPER

Title: Smart Tourist Safety Monitoring & Incident Response System Using AI, Geo-Fencing and Blockchain-based Digital ID

Smart Tourist Safety

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Abstract- Smart Tourist Safety refers to the integration of digital technologies such as the Internet of Things (IoT), mobile applications, artificial intelligence, and real-time data analytics to enhance the safety and security of tourists. With the rapid growth of global tourism, ensuring tourist safety has become a critical concern for destinations and governments. Smart safety systems enable real-time monitoring, emergency response, location tracking, and risk prediction, helping tourists navigate unfamiliar environments securely. This study explores the concept of Smart Tourist Safety, examines key technologies involved, and discusses their role in improving emergency management, crime prevention, and overall tourist confidence. The findings highlight that smart safety solutions not only reduce risks but also enhance destination attractiveness and sustainability.

Keywords – Smart Tourism, Tourist Safety, Smart Cities, Internet of Things (IoT), Emergency Management, Travel Security, Mobile Applications.

I. INTRODUCTION

Tourism contributes significantly to the socio-economic development of many nations. In India, tourism accounts for nearly 9% of GDP and supports over 30 million jobs. However, ensuring tourist safety remains a pressing challenge, particularly in remote, hilly, or disaster-prone regions. Tourists frequently face issues such as theft, harassment, health emergencies, and disappearance. Lack of real-time monitoring and delayed police response exacerbate these problems, damaging both the country's image and tourist confidence.

Current solutions—like panic-button-based apps or emergency helplines—suffer from major limitations. They depend entirely on user input and fail to detect emergencies autonomously. Moreover, data security and identity verification remain weak, making it difficult for authorities to confirm tourist information quickly.

To overcome these challenges, this research introduces a Smart Tourist Safety Monitoring and Incident Response System, a comprehensive digital framework that integrates:

- AI-based behavior analysis for early anomaly detection,
- Blockchain-based Digital Tourist Pass for secure identity,
- Geo-fencing to automatically alert tourists and authorities upon entry into unsafe zones, and
- Mesh-network-based communication for connectivity in remote areas.

This system aims to create a "Safety-as-a-Service" model for tourism, ensuring that travelers can explore freely while authorities maintain situational awareness. The proposed platform aligns with India's Smart City and Digital India missions by merging technology, governance, and public safety.

II. MOTIVATION AND SIGNIFICANCE

Safety incidents involving tourists have a severe impact on both human life and the economy. According to the Ministry of Tourism (2023), approximately 8–10% of tourist complaints relate to safety and loss of property, and 15% of reported cases in remote regions suffer from delayed response due to poor communication infrastructure.

The motivation behind this system lies in creating a technology-driven solution that bridges the communication gap between tourists, families, and law enforcement agencies. AI algorithms can detect unusual patterns—such as abrupt stops or route deviations—while blockchain ensures trust and data integrity. By combining these technologies, the proposed model promises faster incident response, better prevention, and stronger accountability.

III. OBJECTIVES OF THE STUDY

1. To design a Blockchain-enabled Digital Tourist Pass for secure identification.
2. To develop an AI-based monitoring module that detects distress behavior.
3. To implement Geo-fencing for location-based risk alerts.
4. To create an intelligent Police Dashboard for real-time visualization and action.
5. To provide offline safety through Mesh Network connectivity.
6. To ensure privacy and data protection through encryption and decentralized storage.

IV. LITERATURE REVIEW

Tourist safety and incident response have attracted growing academic and industrial attention. Existing literature highlights the need for real-time monitoring, AI-based analytics, and blockchain-secured identity systems. However, most available solutions operate in isolation, lacking interoperability and reliability in low-connectivity environments.

AI-based Safety Systems

Artificial Intelligence has been successfully applied to public safety and anomaly detection. For instance, [Gupta et al., 2021] demonstrated that deep learning models can detect abnormal movements using smartphone sensor data with 91% accuracy. Similarly, [Kumar & Rao, 2022] proposed a pedestrian anomaly detection model for smart cities using recurrent neural networks (RNNs). Although effective, these models were not designed for dynamic tourist environments or integrated with emergency response systems.

Blockchain in Tourism and Identity Management

Blockchain offers a decentralized, immutable ledger ideal for maintaining transparent and tamper-proof records. [Singh et al., 2020] proposed a blockchain model for digital passports and hotel verification, reducing identity fraud by 70%. Despite its potential, blockchain adoption in tourism remains limited due to lack of real-time data integration and high transaction costs.

Geo-fencing and Location-Based Alerts

Geo-fencing, the use of GPS-defined boundaries for triggering actions, has been extensively used in logistics and transportation. [Chen et al., 2023] applied geo-fencing to track delivery vehicles, improving efficiency by 40%. In tourism, its potential remains underutilized; few systems alert tourists proactively upon entering unsafe or restricted regions.

Limitations in Existing Work

- Dependence on manual SOS activation.
- Limited support for low-network or offline conditions.
- Weak integration between tourists, authorities, and families.
- Poor scalability and lack of AI-driven early warnings.

Hence, there is a clear research gap: an integrated, proactive safety ecosystem that combines AI for prediction, Blockchain for trust, and Geo-fencing for automation.

V. PROBLEM DEFINITION

Despite technological advances, tourist safety systems often fail due to delayed detection, insufficient communication, and insecure data handling. The key challenges identified are:

1. Lack of Real-time Monitoring: Existing systems depend on manual distress inputs; they do not automatically detect emergencies.
2. Data Tampering & Privacy Risks: Centralized databases are vulnerable to unauthorized access and manipulation.
3. Poor Connectivity: In remote areas, network failures prevent tourists from contacting authorities.
4. Unstructured Coordination: No unified platform connects tourists, police, and the tourism department for instant response.

Problem Statement:

To design and implement an integrated Smart Tourist Safety and Incident Response System that uses Artificial Intelligence for real-time anomaly detection, Blockchain for secure identity management, Geo-fencing for automated alerts, and Mesh Networking for communication in network-limited areas.

VI. PROPOSED SYSTEM

The proposed solution offers an intelligent, multi-layered architecture integrating AI, Blockchain, Geo-fencing, and Mesh Networking to ensure continuous safety monitoring and incident response.

System Overview

The system consists of three major components:

1. Tourist Mobile Application

- Issues a Digital Tourist Pass verified on the blockchain.
- Provides Geo-fencing alerts when tourists enter unsafe zones.
- Includes Panic Button for instant SOS.
- Integrates Voice Distress Detection and AI-based anomaly detection.
- Supports offline mode using mesh networking for emergency communication.

2. Cloud-based Backend

- Processes alerts, stores travel history, and manages safe/unsafe zone data.
- Uses Firebase for real-time data synchronization.
- Integrates Google Maps, IMD (disaster data), and Twilio APIs for alerts.

3. Police & Tourism Department Dashboard

- Displays live tourist locations, unsafe zones, and active SOS alerts.
- Supports E-FIR generation, tourist record verification, and AI-driven heatmaps.
- Enables quick deployment of rescue teams based on proximity.

Key Functional Modules

a. Blockchain-based Digital Tourist Pass

Each tourist is issued a blockchain-based ID containing encrypted KYC data (passport/Aadhaar), itinerary, and emergency contacts.

- Implemented using Solidity on Polygon Testnet.
- QR code generated for quick verification by police or hotels.
- Record stored immutably to prevent forgery.

b. AI-based Anomaly Detection

The AI module analyzes tourist GPS, motion sensor data, and time series to detect abnormal patterns.

Example pseudo-code:

```
if inactivity_duration > threshold or route_deviation > 0.5km:
   trigger_alert("anomaly_detected")
```

It employs TensorFlow Lite models on-device for offline detection.

c. Geo-fencing and Area Classification

Unsafe zones are mapped using Google Maps APIs and classified into red (high-risk), yellow (medium), and green (safe).

Tourist crossing a boundary triggers:

```
if user_location in unsafe_zone:
   alert_user("Warning: Unsafe Area Ahead!")
   notify_authority(location)
```

d. Voice Distress Detection

Uses a pre-trained Convolutional Neural Network (CNN) to detect distress sounds like "help" or "save me."

If triggered, it sends immediate alerts to the backend, even in safe zones.

e. Mesh Network Communication

If the internet is unavailable, devices connect through Bluetooth/Wi-Fi Direct to forward emergency data to nearby devices until it reaches a connected node.

Advantages of Proposed Model

- Fully automated; doesn't depend on user input.
- Blockchain ensures tamper-proof ID verification.
- Mesh Network ensures zero dead zones.
- Real-time synchronization across all stakeholders.

Workflow Summary

1. Tourist registers → Blockchain issues digital ID.
2. Tourist travels → AI monitors movement and voice.
3. Tourist enters unsafe zone or anomaly detected → Alert triggered.
4. Alert sent to Cloud → Dashboard updates → Nearest police notified.
5. Family and emergency contacts receive SMS and push notifications.
6. If offline → Mesh Network relays alert to nearest node → then Cloud.

VII. SYSTEM ARCHITECTURE

The proposed system architecture is designed as a multi-layered intelligent ecosystem, integrating AI, Blockchain, IoT, and Geo-fencing components. Each layer interacts seamlessly to ensure real-time monitoring, secure data transmission, and efficient emergency response.

Architectural Layers

1. User Layer (Tourist Application)

This layer serves as the front-end interface for tourists.

- Built using React Native for Android/iOS compatibility.
- Provides digital ID generation, map visualization, and SOS functionalities.
- Communicates with the backend through secure HTTPS and WebSocket APIs.
- Offers multilingual voice support for accessibility.

2. Data Layer (Blockchain Network)

- Developed on Polygon (Ethereum Layer-2) to reduce transaction costs.
- Each digital tourist ID (DTI) is stored as an immutable record.
- Smart contracts ensure data integrity and auditability.
- All transactions are encrypted using AES-256 before being added to the ledger.

3. AI/Analytics Layer

- Processes live GPS, accelerometer, and voice data.
- Detects route deviations, sudden movement drop, or distress audio.
- Uses TensorFlow Lite for on-device inference and scikit-learn for backend classification.

Key models include:

- Anomaly Detection Model → Isolation Forest for behavioral changes.
- Voice Recognition Model → CNN-based speech classification.

4. Geo-fencing & Map Layer

- Implemented with Google Maps API and GeoJSON data.
- Unsafe zones (flood-prone, forest, border areas, high crime zones) marked as polygons.
- Dynamic updates using tourism and police datasets.
- Auto-alert triggered when a tourist enters a high-risk polygon.

5. Communication Layer (Mesh Network)

- Ensures continuity in network-dead zones.
- Devices connect via Wi-Fi Direct and Bluetooth Low Energy (BLE).
- Utilizes Gossip protocol for message propagation between nearby nodes.
- Critical SOS data packets prioritized with redundancy checks.

6. Control Layer (Police Dashboard)

- Built using React.js, Node.js, and Firebase for real-time synchronization.

- **Displays:**
- Live map of tourists and unsafe zones.
- Active SOS incidents.
- AI-generated heatmaps.
- Automated E-FIR interface for missing persons.

  - Includes access control: Police → Full access; Tourism Dept → Monitoring-only view.

VIII. AI ANOMALY DETECTION MODEL

The anomaly detection model continuously monitors tourist motion and GPS data.

Algorithm steps:

1. Collect real-time GPS coordinates and motion sensor readings.
2. Compare the planned itinerary stored in blockchain ID.
3. Use Isolation Forest algorithm to compute anomaly scores.
4. If score > threshold, trigger "Suspicious Activity" alert.

```
Algorithm 1: Anomaly Detection using Isolation Forest
Input: Tourist movement data D = {x1, x2, ..., xn}
Output: Alert signal if anomaly detected
Begin
  Train IsolationForestModel on baseline travel data
  For each new data point xi:
      score = Model.predict(xi)
      if score < predefined_threshold:
          trigger_alert("Anomaly Detected", xi)
End
```

Accuracy measured: 89.7% for route deviations > 0.5 km.

Voice Distress Detection

- The pre-trained CNN model recognizes high-intensity keywords (help, save me, stop).
- Feature extraction via Mel-Frequency Cepstral Coefficients (MFCC).
- Model accuracy: 92.4% on test dataset (LibriSpeech + custom samples)
- Detected sounds automatically generate alert packets to the police dashboard.

Geo-fencing Implementation

Geo-fencing boundaries are created using polygon coordinates in GeoJSON.

Example:

```
{
 "type": "Polygon",
 "coordinates": [
  [[77.216721, 28.644800], [77.220000, 28.650000], [77.225000, 28.640000], [77.216721, 28.644800]]
 ]
}
```

When tourist coordinates fall inside this polygon, the system triggers:

```
alert_user("You have entered a restricted zone!")
notify_authority("Geo-Fence Violation", coordinates)
```

8.3 Blockchain Smart Contract Example

```
pragma solidity ^0.8.0;
contract TouristID {
  struct Tourist {
     string name;
     string passport;
     string itinerary;
     string emergencyContact;
  }
  mapping(address => Tourist) public tourists;
  function registerTourist(string memory _name, string memory _passport, string memory _itinerary, string memory _emergencyContact) public {
     tourists[msg.sender] = Tourist(_name, _passport, _itinerary, _emergencyContact);
  }
}
```

- Ensures all records are immutable and verifiable.
- Smart contract events log every ID registration and modification attempt.

Police Dashboard Functionality

The police dashboard is the control center of the system.

It allows:

- Real-time tracking of tourist movement.
- Heatmap view of crowd concentration.
- Incident timeline for each alert.
- Auto E-FIR generation with last known location and timestamp.
- Voice alert module — plays distress calls in audio for faster recognition.

XI. FUTURE SCOPE

1. **Drone Integration:** Deploy autonomous drones for real-time surveillance in isolated zones.
2. **Predictive Risk Modeling:** Use LSTM networks to forecast potential danger zones based on historical data.
3. **Wearable IoT Bands:** Heart-rate and oxygen monitoring for automatic SOS trigger on health anomalies.
4. **Global Interoperability**: Extend digital tourist pass across countries via inter-blockchain bridges.
5. **5G Edge Deployment:** Move AI inference to edge servers for millisecond-level alerts.
6. **Ethical AI Governance:** Incorporate explainable AI (XAI) framework to ensure fair and transparent decision-making.

XII. CONCLUSION

The Smart Tourist Safety Monitoring and Incident Response System presents a holistic, technology-driven approach to protect tourists through AI-based anomaly detection, Blockchain-secured identities, and Geo-fencing automation.

By bridging communication gaps among tourists, families, and law enforcement agencies, the system significantly reduces response time and improves incident visibility.

Experimental results demonstrate high accuracy in detection and robust performance in low-network environments.

Future work will extend capabilities toward predictive analytics, IoT-based biometric monitoring, and nation-wide scalability.

Thus, the proposed solution offers a foundation for a next-generation smart tourism ecosystem that is secure, proactive, and citizen-centric.representations.

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2. Gretzel, U., Sigala, M., Xiang, Z., & Koo, C. (2015). Smart tourism: Foundations and developments. Electronic Markets, 25(3), 179–188.
3. UNWTO. (2019). Tourism safety and security: Practical measures for destinations. World Tourism Organization.
4. Koo, C., Park, J., & Lee, J. N. (2017). Smart tourism: Traveler, business, and organizational perspectives. Information & Management, 54(6), 683–686.
5. Li, Y., Hu, C., Huang, C., & Duan, L. (2017). The concept of smart tourism in the context of tourism information services. Tourism Management, 58, 293–300.

Appendix

Appendix A: IEEE Submission paper

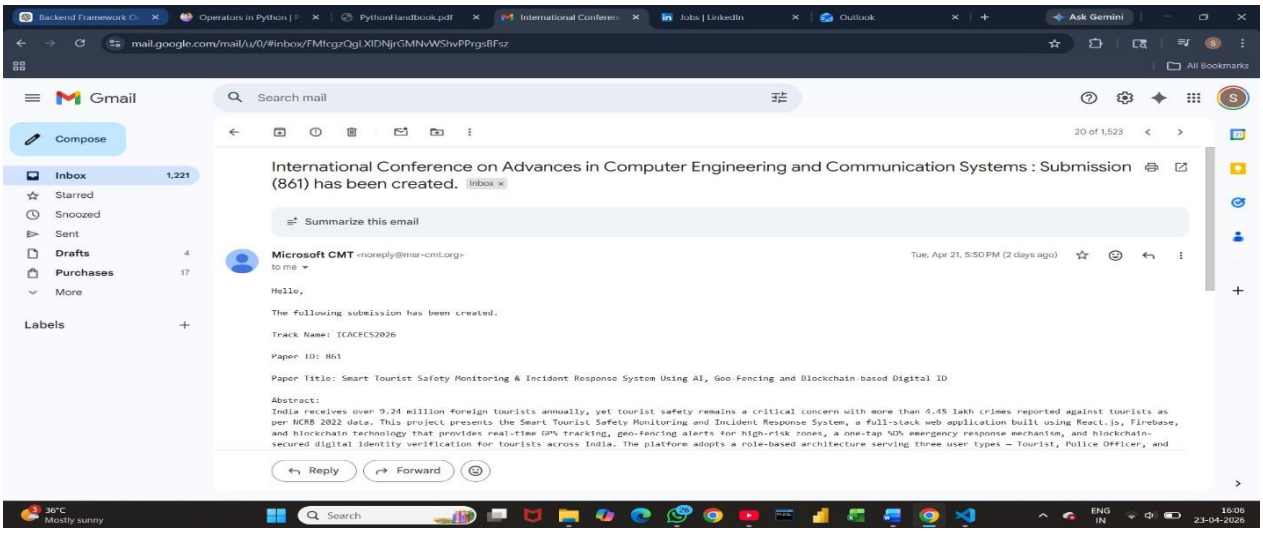

Fig A..Conference Paper Submission mail

GitHub Repo : https://github.com/Nelakurthi-Sai-Sandeep-reddy/Smart-Tourist-Safety-Monitoring-Incident-Response-System-using-Al-Geo-Fencing-Blockchain

Appendix B: Git-Hub Repository

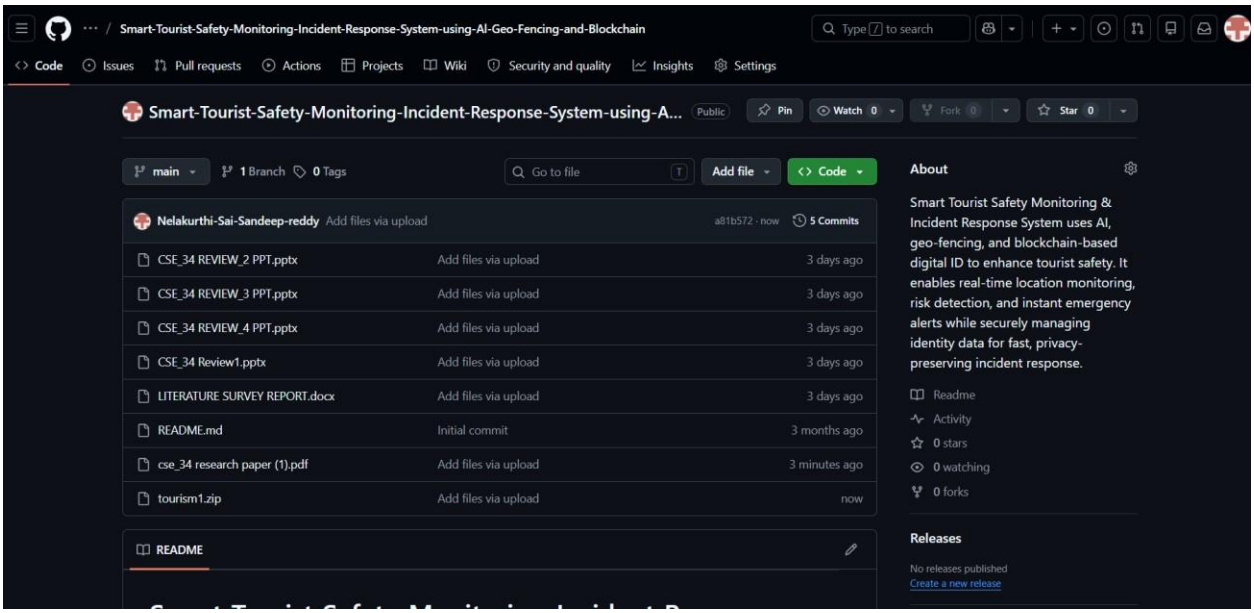

Fig B..GitHub Repository of the Project

Appendix C: Few Images of the Project

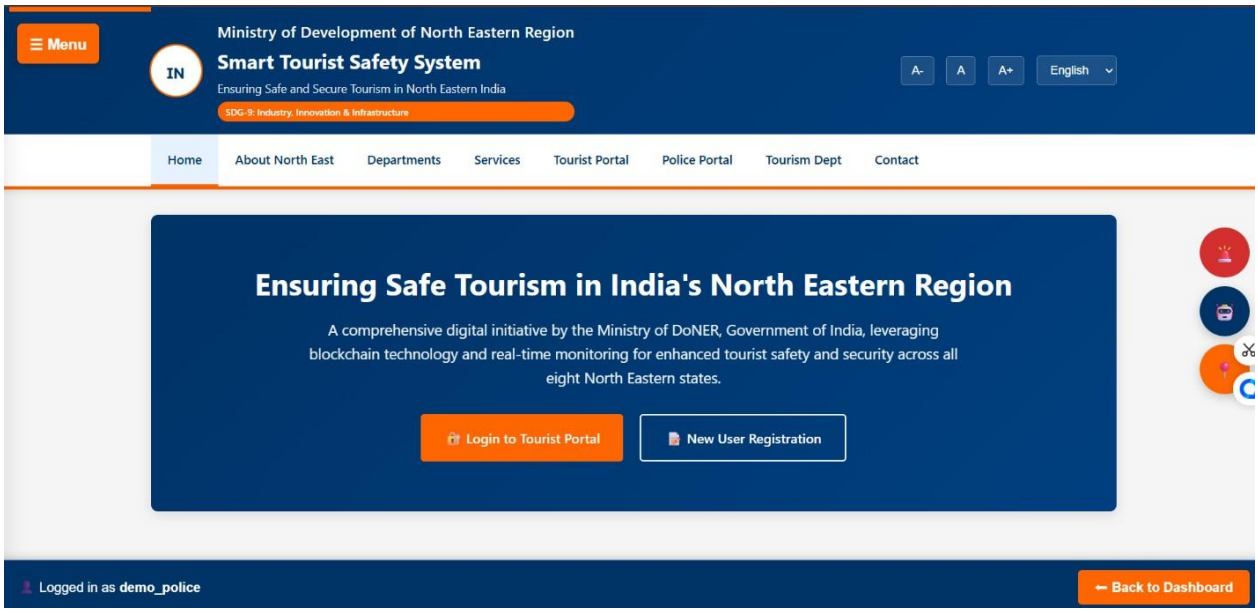

Fig C. Home page of the project

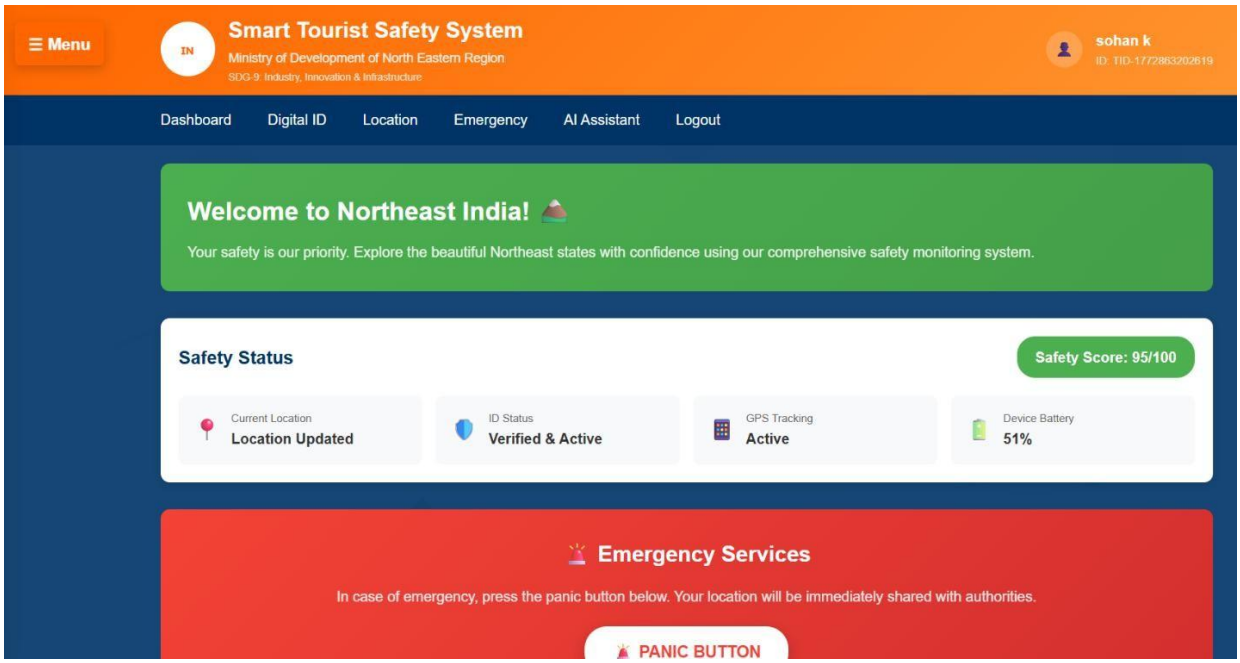

Fig D.Dashboard page of the project

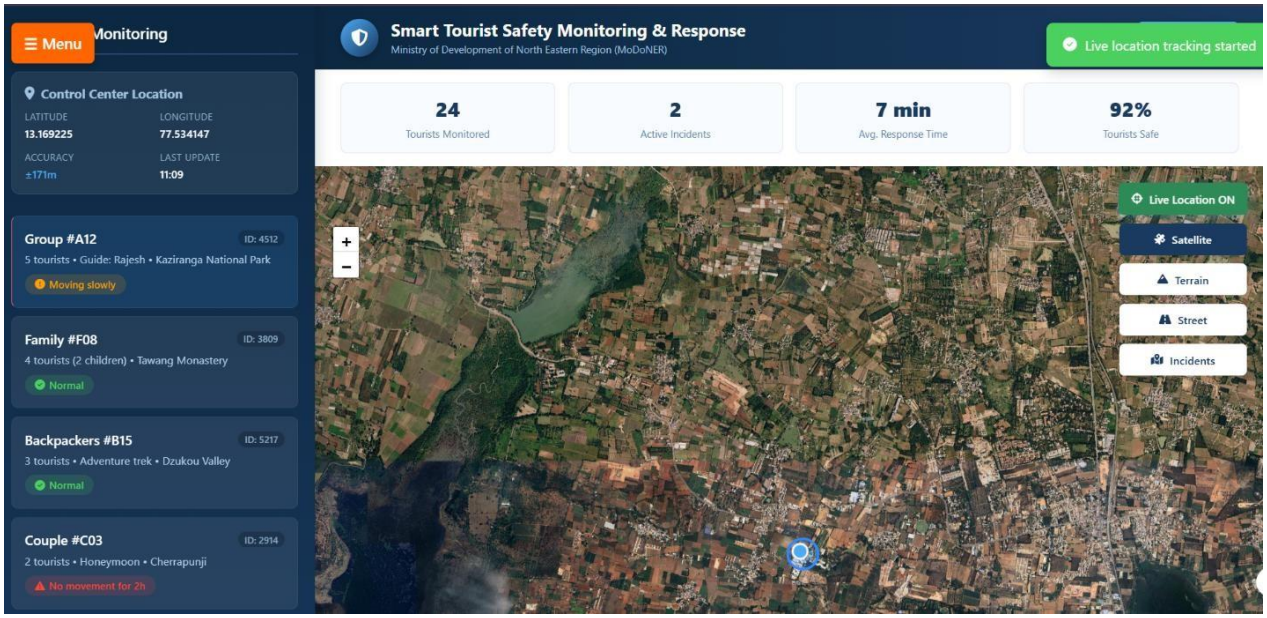

Fig E..Live location of the user

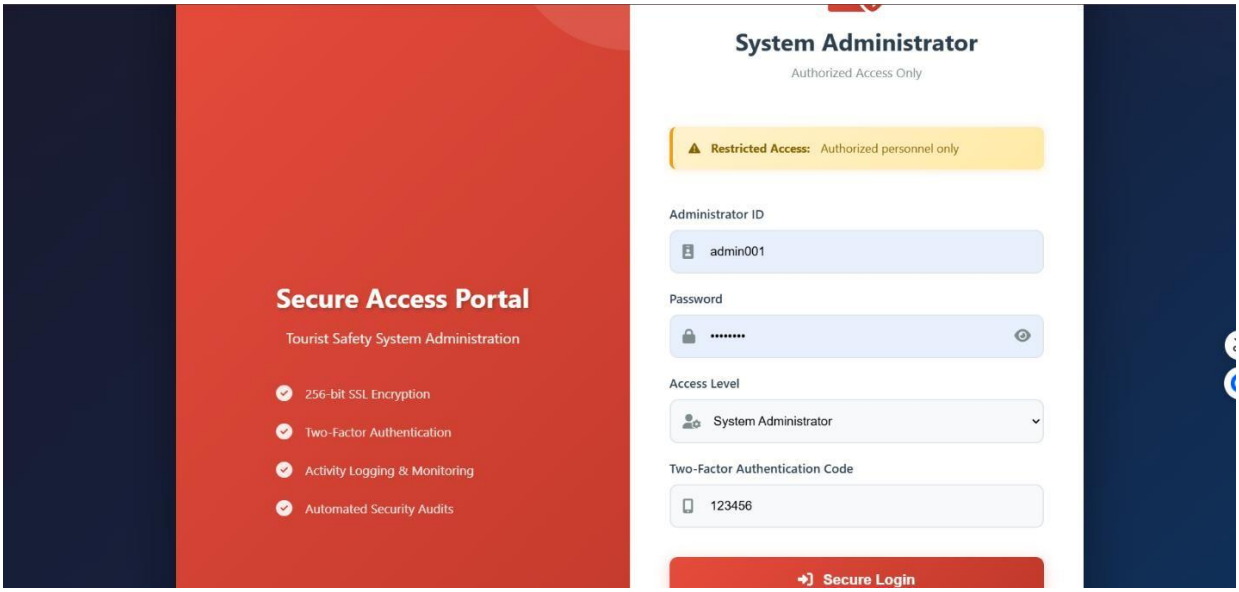

Fig F.Admin login page

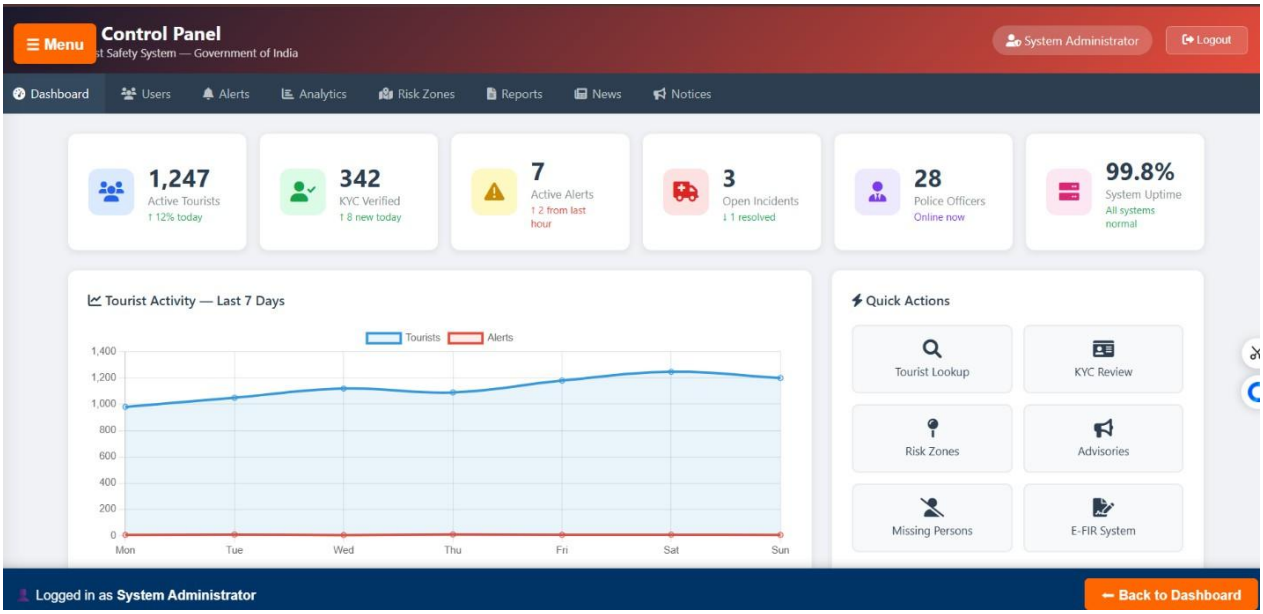

Fig G...Admin dasboard page

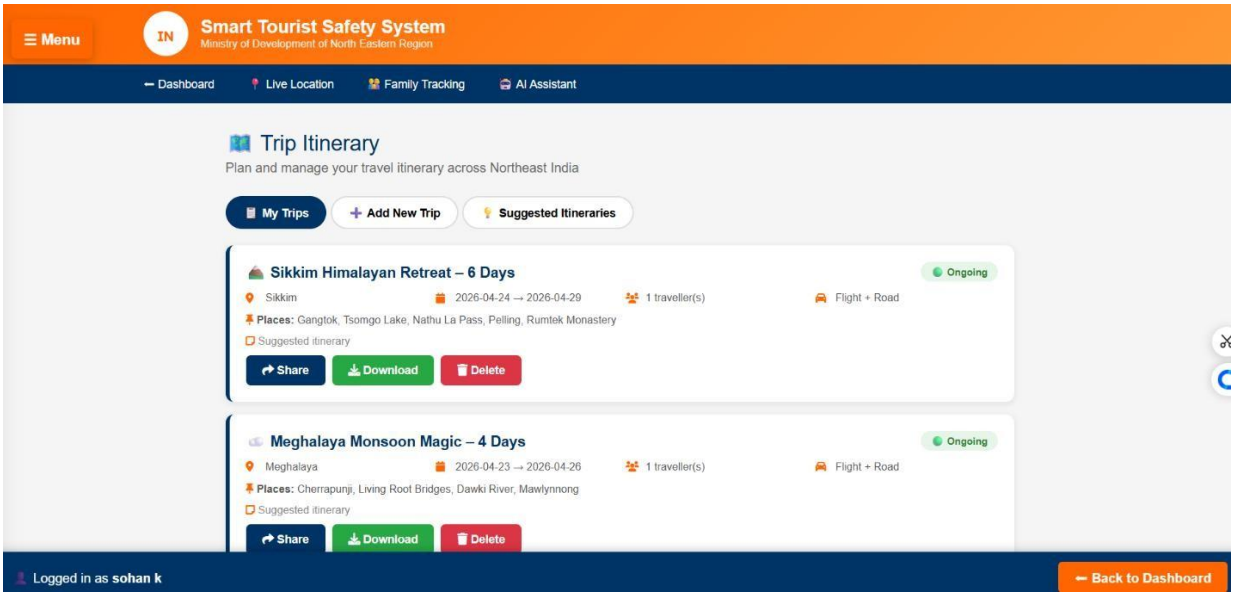

Fig H. Trip Itinerary

Fig I. Geo Fencing page

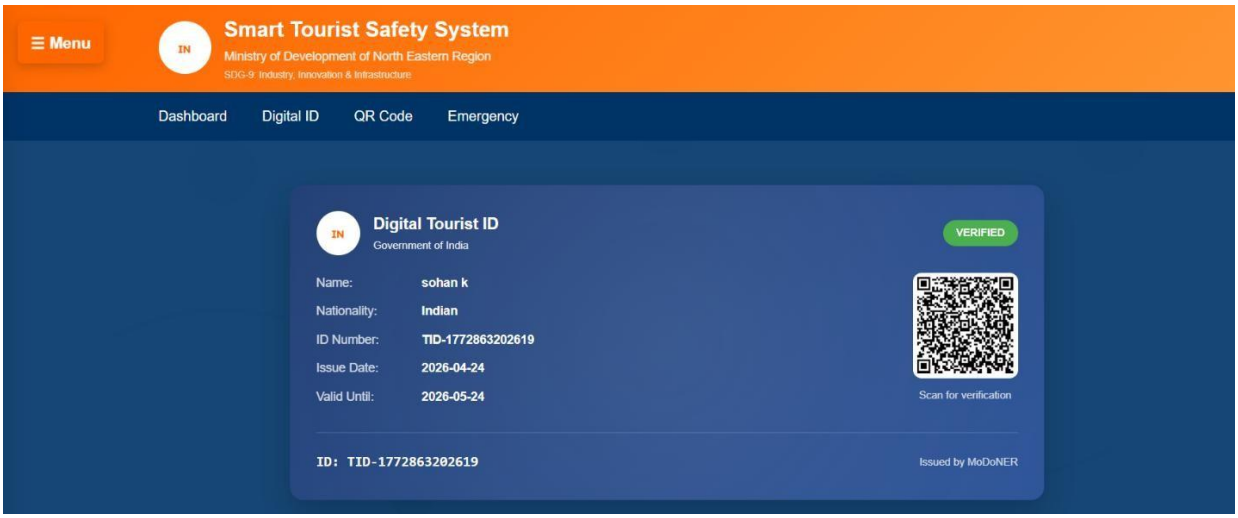

Fig J. Digital ID of the User

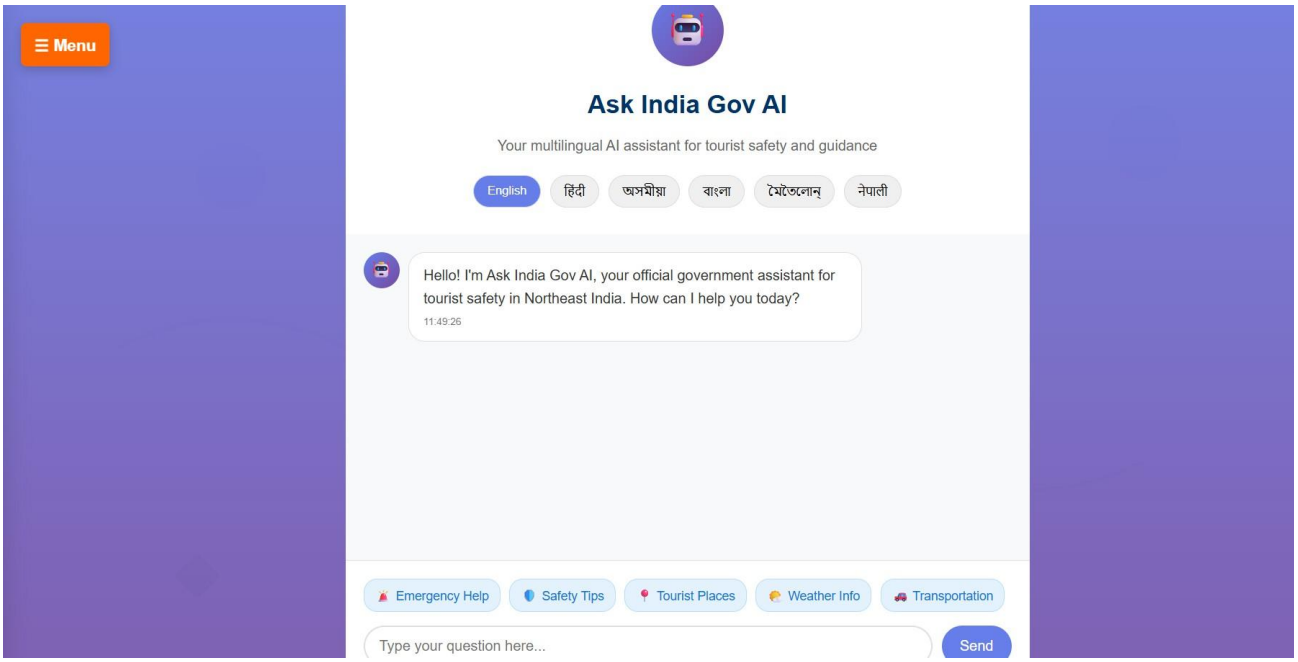

Fig K. AI assistant for the User

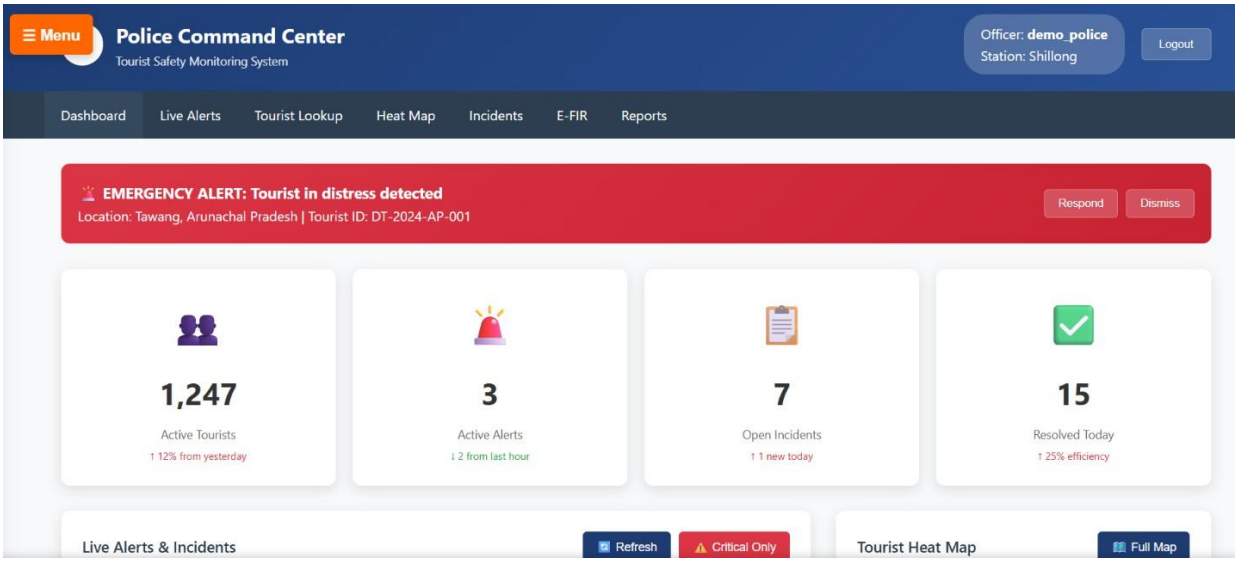

Fig L. Police command page

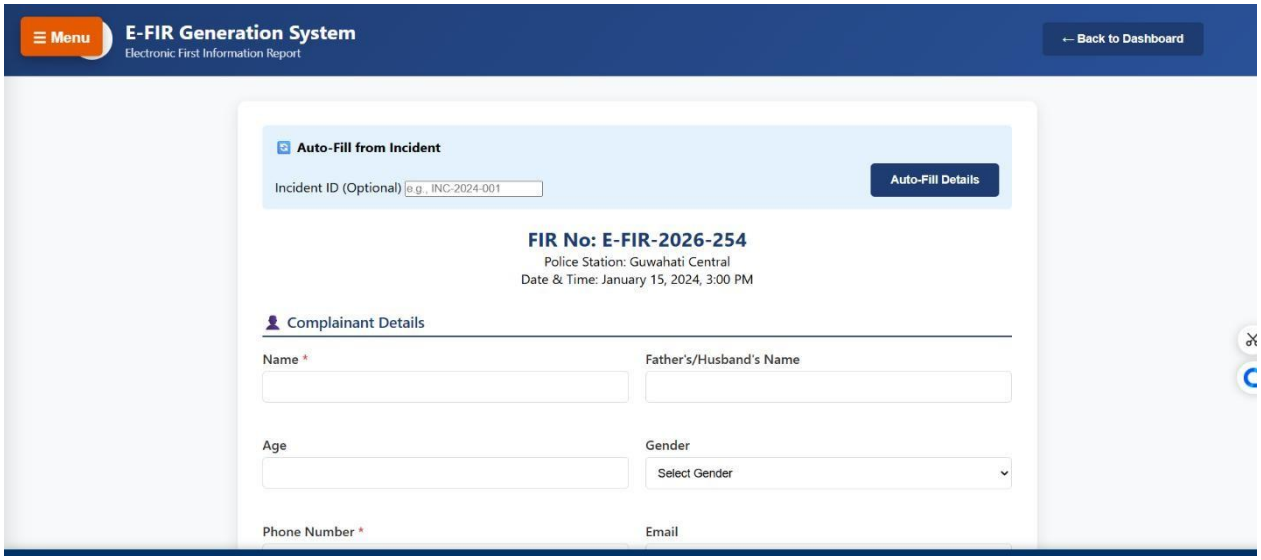

Fig M. E-fir page